

The following pages report the document:

A. Amicarelli, 2025; RANS SPH CFD for Air Quality in and around Urban Canopies; paper preprint; protocol RS2502, pp.1-27.

# RANS SPH CFD for Air Quality in and around Urban Canopies

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## Abstract (within 150 words)

A RANS CFD-SPH code is validated on the Urban Canopy (UC) of a wind-tunnel experiment in terms of mean concentration. Despite focusing on the most challenging 25% of the Canopy, the code satisfies the state-of-the-art thresholds of Chang & Hanna (2004, [1]) for the validation metrics and shows an above-coverage performance vs. 28 state-of-the-art deterministic codes. The application of a standard RANS Wall Function (WF) seems incoherent with the erosion of the Neutral Surface Boundary Layer (NSBL) in the wake-interference regime of UCs, causing relevant pollutant under-diffusion with Ground-Level Sources. A Smoothed WF is proposed to switch progressively from the NSBL WF to no WF with the node-to-wall distance. The Fractional Bias magnitude and the Normalised Mean Square Error are reduced of 65% and 78% by the Smoothed WF (80%) and  $Sc_T$ -TGS closure (20%, without ballistic regime). A full-scale accidental release of hydrocarbons from a gas station involves 36 buildings.

## Keywords (5–10)

RANS CFD; SPH; Urban Canopy; Wall Functions; Smoothed Wall Function;  $Sc_T$ -TGS turbulence closure; CFD-OUT-sweepPDF; Air Quality; Turbulent Schmidt number; Coefficient of turbulent diffusion.

## 1. INTRODUCTION

Within the scope of Outdoor Air Quality OAQ, an Urban Canopy UC can be defined as a set of non-isolated buildings forming a number of connected street canyons. CFD modelling represents the most accurate numerical approach to simulate OAQ in urban areas at spatial resolutions smaller than 2km (i.e. the micro-scale) offering powerful tools to carry out ex post, risk and back-cast analyses on the impact of OAQ on human health, the environment and the public and private goods targeted. A non-exhaustive state of the art on CFD modelling for OAQ in and around UCs is described hereafter. The associated studies simulated buildings as explicit obstacles and presented validations on concentration statistics. Where available, the values of the turbulent Schmidt number  $Sc_T$ , i.e. the ratio between the turbulent viscosity and the coefficient of turbulent diffusion, are reported.

Li et al. (2006, [2]) described a detailed revision on CFD models for OAQ within a single street canyon stating that there was no CFD validation study at that time concerning OAQ with UCs. Hsieh et al. (2007, [3];  $Sc_T=0.63$ ) validated the concentration mean and variance, estimated by a deterministic Reynolds-Averaged Navier-Stokes (RANS) model based on the Finite Volume Method (FVM), by comparison with the water-channel Mock Urban Setting Test (MUST) experiment (Hilderman & Chong, 2007, [4]; scale ratio 1:205). This is the water similarity variant of the open-field MUST experiment of Yee & Biltoft (2004, [5]; scale ratio 1:10). The latter reproduced the dispersion of a passive pollutant around a matrix of buildings, which were misaligned with the free stream and formed a virtual urban area in Outdoor atmosphere. Milliez & Carissimo (2007, [6]) and Milliez & Carissimo (2008, [7]) tested a RANS k-eps baroclinic model for the airflow on the open-field MUST experiment in terms of concentration mean and variance. Shi et al. (2008, [8]) validated a Large Eddy Simulation (LES) model under the experimental turbulent flow of Davidson et al. (1996, [9]) with the dispersion of a passive pollutant around a matrix of obstacles aligned with the free stream. The Sub-Grid Scheme (SGS) for the airflow was either Smagorinsky model or an SGS dynamic model; the first was combined with the SGS turbulent Schmidt number  $Sc_{T,\Delta}=0.7-0.8$ , the latter with  $Sc_{T,\Delta}=0.7$ ; no SGS model seems available for the mean concentration. Efthimiou et al. (2011, [10]) tested a deterministic RANS FVM code on the open-field MUST experiment, in terms of concentration mean and variance. Leuzzi et al. (2012, [11]) validated the RANS-LSM code LAGFLUM on the wind-tunnel MUST experiment (Bezpalcova, 2007, [12]), i.e. the wind-tunnel

variant (scale ratio 1:750) of the homonymous open-field experiment. LSMs stands for ‘Lagrangian Stochastic Models’ (e.g. Thomson, 1987, [13]; Gariazzo, 2004, [14]). They solve a vector Stochastic Differential Equation (SDE; e.g. Morkisz & Przybylowicz, 2017, [15]) for the instantaneous velocity, whose statistics are input data, to assess the concentration fields. Validations concerned the concentration mean, variance and skewness. The input velocity statistics were provided either by a diagnostic micro-scale meteorological model (Amicarelli et al., 2011, [16]) or by the deterministic RANS-FVM code MISKAM, described in Eichhorn & Kniffka (2010, [17]). A precursor validation on the same topic, without either MISKAM data or concentration skewness, was reported in Amicarelli et al. (2011, [16]). Tominaga & Stathopoulos (2012, [18]) tested a deterministic RANS RNG (Renormalisation Group) k-eps model and a deterministic LES-SGS model on their experimental turbulent flow with the dispersion of a passive pollutant around a matrix of buildings. Tominaga et al. (2013, [19]) validated 6 RANS and 2 LES CFD mesh-based models on the wind-tunnel MUST experiment, offering valuable inter-comparisons with no model tuning. Together with the positive validation metrics for all the models, they highlighted a systematic and relevant under-diffusion of the pollutant mass associated with the overestimation in the RANS mean concentration along the plume centreline, with respect to the measurements and the LES models. Kellnerová et al. (2018, [20]) used a LES code based on the Finite Difference Method (FDM) to validate the LES-filtered concentration in a street canyon with pitched roofs, by comparison with previous measurements. Bahlali et al. (2019, [21]) tested a CFD code on the open-field MUST experiment in terms of mean concentration. Four approaches were validated: a deterministic RANS FVM model for airflow dynamic and pollutant transport, with either k-eps or RS-eps (Reynolds’ Stress tensor-eps) closure; a RANS-Probability Density Function (RANS-PDF) Particle-in-Cell (PiC) scheme coupled with either the aforementioned models for the airflow. Unlike more common applications, the RANS-PDF model was used as a LSM, being the velocity statistics taken in input from the deterministic model, besides the mean-pressure gradient.  $Sc_T$  was declared in the range 0.7-1, without more specific details on the values used in each simulation. Chaloupecká et al. (2019, [22]) presented a puff model for UC with a simplified estimate of the concentration turbulent fluctuations around the mean, following an experimental study under UC focused on the assessment of the release time of intermittent puffs (Chaloupecká et al., 2018, [23]). Papp et al. (2021, [24]) analysed the effects on OAQ of inhomogeneity in the building height and of stagger in the building pattern via experiments and LES; a following experimental study integrated the concentration turbulent fluctuations around the mean (Papp et al., 2024, [25]). Hadziabdic et al. (2022, [26]) validated the in-house code T-Flows (<https://github.com/DelNov/T-Flows>), developed at TU Delft by Niceno (2001, [27]) and Niceno & Hanjalic (2005, [28]), on the water-channel MUST experiment. They considered either a hybrid Unsteady RANS (URANS)-LES numerical solution or a stand-alone URANS approach, both with  $Sc_T=0.9$ . Zhang et al. (2024, [29]) analysed the impact of building layout on pollutant transport into an industrial park via CFD simulations and field monitoring, depending on atmospheric stability conditions. Besides the wind-tunnel MUST experiment, which is a reference for several CFD validations, other OAQ scale physical models with UC were carried out at the Environmental Wind Tunnel Laboratory (EWTL) of the University of Hamburg, such as the Joint Urban 2003 experiment (Klein & Young, 2011, [30]; Klein et al., 2011, [31]).

In the state of the art above, a Lagrangian deterministic RANS CFD-SPH code was developed to simulate turbulent flows with pollutant transport for Outdoor/Indoor/Duct AQ applications (Amicarelli, 2024, [32]; Amicarelli, 2025, [33]). The code is based on Kato & Launder’s improved k-eps (Kat&Lau-k-eps) turbulence closure of the Momentum Equation (ME) and the  $Sc_T$ -TGS turbulence closure of the Balance Equation (BE) for the mean concentration (Cm-BE). The Smoothed Particle Hydrodynamics (SPH) numerical method is meshless, directly reproduces the interfaces between fluids and phases, and permits to represent the mean advection terms of the BEs freely and exactly. SPH is suitable for scheme couplings and multi-physics, and is prone to general-purpose CFD modelling. Explicit time integration of independent particle Partial Differential Equations (PDEs) is allowed by the Weakly Compressible (WC) approach. The aforementioned code was

validated under grid turbulence, a shear flow and a confined flow past a wall-mounted cubic obstacle. The aims of the current study are to validate such a method in assessing the mean concentration in and around a complex UC, to fix any associated shortcoming of general interest by improving the mathematical/numerical models where needed, and to compare the  $Sc_T$ -TGS closure with the state-of-the-art approach of constant  $Sc_T$  ( $Sc_T$ -const) under a UC flow. The present paper is organised as follows. The new code version CFD-OUT-sweepPDF v.3.0.0 is synthetically described (Sec.2). Its main novelty is a Smoothed Wall Function for Obstacle Canopies (OCs), i.e. sets of non-isolated obstacles like a UC (Sec.3). The code is validated on the wind-tunnel MUST experiment (Bezpalcova, 2007, [12]) as an application to OAQ with UC, a series of model inter-comparisons is reported and an associated full-scale application is shown (Sec.4). The results and novelties of the current study are presented in the conclusions (Sec.5).

## 2. THE CODE CFD-OUT-SWEEPPDF V.3.0.0

CFD-OUT-sweepPDF v.3.0.0 (Amicarelli et al.) is a deterministic Lagrangian RANS CFD-SPH code for turbulent flows with pollutant transport. The following applications have been investigated: OAQ, Indoor Air Quality (IAQ) and air-treatment processes for Duct Air Quality (DAQ). They include ordinary stack emissions and accidental releases. The code reproduces any useful spatial resolution under a maximum spatial cover of  $2\text{km} \times 2\text{km} \times 0.5\text{km}$ . Both non-stationary and stationary conditions are represented.

Analogously to the WC approach for Implicit LES-SPH (ILES-SPH; e.g. Cui et al., 2019, [34]; Cercos-Pita et al., 2017, [35]) and Monotone Integrated LES-SPH (MILES-SPH, e.g. Marongiu et al. 2010, [36]; Gu et al., 2017, [37]; Koukouvinis et al., 2018, [38]), the SPH RANS Continuity Equation (RANS-CE) of the code computes the mean pressure via a barotropic Equation of State (EoS), averaged after Reynolds. Unlike the state-of-the-art SPH models that merge turbulent with physical diffusion in the CE via advanced techniques (e.g. Jeong & Lee, 2025, [39]; Isik & He, 2023, [40]), as demonstrated in Mayrhofer et al. (2013, [41]), the RANS-CE of the present code features an explicit turbulent transport term, homologous to the LES-CE sub-grid term of Sun et al. (2019, [42]). The code relies on Kato & Launder’s improved k-eps closure (Kat&Lau-k-eps). Among the 2-equation 1st-order turbulence closures for the Momentum Equation (e.g. Fang et al., 2017, [43]), Kat&Lau-k-eps is deemed as one of the most accurate. The code features the ‘ $Sc_T$ -TGS’ turbulence closure for the mean-concentration Balance Equation (BE); it is coherent with Taylor’s theory (Taylor, 1922, [44]), grid turbulence and shear flows, and reproduces both the effects of velocity autocorrelation (e.g. Fang et al., 2022, [45]) and the dependence of  $Sc_T$  on the turbulence Reynolds number  $Re_T$ . The semi-implicit Euler’s scheme and several hybrid forward-backward Euler’s schemes drive time integration. To enhance the stability domains the code features a spatial linear filter in RANS-CE, ruled by a RANS-SPH smoothing coefficient to inhibit the SPH zero-energy modes (e.g. Vignjevic et al., 2000, [46]) associated with variable co-location, a stabilisation term for the Momentum Equation (ME) and a shell SPH filter. The ME stabilisation term has the same aim as the SPH Particle Shifting Techniques (e.g. Crespo et al, 2008, [47]; Aristodemo et al., 2015, [48]; Cui et al., 2020, [49]) of reducing the deformations of the control volumes (Fang et al., 2017, [50]) and also inhibits the so-called SPH tensile instability (e.g. Zhang et al., 2017, [51]) thanks to its numerical limiter. The SPH Time-Space Ensemble (SPH-TSE) filter improves the accuracy of the ensemble statistics under specific conditions. The SPH filter applies to all the derivatives in the BEs, and to the functions filtered at the monitoring points and at the barycentres of the positioning grid. The SPH filters of a generic function ( $C_1$ -consistent similarly to Liu & Liu, 2006, [52], also at boundaries), gradient ( $C_1$ -consistent at boundaries) and ‘quasi-Laplacian’ (i.e. the divergence of the product of the function gradient by a scalar) consider contributions from those fluid particles, those solid particles of rigid solid bodies and those elements of fixed walls that are located in the SPH neighbourhood of a computational particle. A kernel-based anti-cluster strategy is used for fluid-fluid and fluid-body interactions. A unified anti-cluster and anti-penetration strategy applies to fluid-wall interactions, based on reactions as numerical collisions. A series of spatial reconstruction schemes applies,

included RANS SPH Wall Functions (WFs) and the Sub-Particle Pollutant Source (SPPS) scheme. WFs depend on the three hydraulic-roughness regimes and the depths of the turbulent Boundary Layer (BL) and of the Neutral Surface BL (NSBL). The BE of the mean Turbulent Kinetic Energy (TKE-BE) is adapted to the WF of its dissipation rate. Under fluid flows confined by solid rigid boundaries, the conservative BE for the particle Control Volume (CV-BE) can be activated to conserve the fluid global volume exactly. The peak value of the compensated velocity Lagrangian structure function is assessed to compute  $Re_T$  and the Lagrangian integral time scale of turbulence  $T_L$ , which are needed by  $Sc_T$  and the SPH-TSE filter. A sole set of computational nodes is used to simulate both the airflow and the pollutant transport. The code treats fixed and mobile solid boundaries of any complex shape, switches from RANS to ILES or Navier-Stokes' BEs, and can be potentially used as a general-purpose CFD code. For instance, SPH models have been widely applied to free-surface and multi-phase flows (Le Touzé & Colagrossi, 2025, [53]), multi-physics phenomena (e.g. Dominguez et al., 2022, [54]; Isik & He, 2023, [55]), included hydroelastic Fluid-Structure Interactions (Khayyer et al., 2021, [56]), and non-Newtonian flows (e.g. Shao & Lo, 2003, [57]; Mascara et al., 2024, [58]). A part of the contributions of the fixed walls to the SPH filters is represented by geometrical quadrature formulae over single fixed-wall elements. Newton-Euler's equations rule Solid Dynamics for rigid solid bodies. Unlike many state-of-the-art formulations used in 3D, the mathematical model for body rotation is fully 3D, as the homologous model of Koukouvinis & Anagnostopoulos (2023, [59]) that is integrated into a mesh-based CFD code to transport fish bodies in a hydraulic turbine via the technique of immersed boundaries.

The main parts of those source lines of the current code dealing with geometrical quadrature formulae over single fixed-wall elements are imported from the code SPHERA v.10.0.0 (RSE SpA, 2022, [60]) and were presented in Di Monaco et al. (2011, [61]). The main parts of those source lines of the current code dealing with the two BEs for Computational Solid Dynamics are imported from the code SPHERA v.10.0.0 (RSE SpA, 2022, [60]) and were presented in Amicarelli et al. (2015, [62]), Amicarelli et al. (2020, [63]), Amicarelli et al. (2021, [64]) and Amicarelli et al. (2022, [65]).

The mathematical/numerical models, the pre/post-processing procedures and the results of the previous versions of the code are available in Amicarelli (2024, [32]) and Amicarelli (2025, [33]). The main novelty introduced in the current version is a Smoothed Wall Function for the mean-TKE dissipation rate under the wake-interference regime of OCs (Sec.3). The code is written in Fortran 95 and has not been distributed.

### 3. A SMOOTHED WALL FUNCTION FOR THE DISSIPATION RATE OF THE MEAN TKE IN OBSTACLE CANOPIES

The Wall Function WF for the dissipation rate of the mean TKE  $\bar{\varepsilon}$  replaces the value computed by the SPH BE for  $\bar{\varepsilon}$  (eps-BE), whose diffusion term is the object of relevant errors, due to the noticeable gradient of  $\bar{\varepsilon}$  close to walls. The systematic application of a standard WF for  $\bar{\varepsilon}$  to the nodes of a RANS simulation seems incoherent with the erosion of the NSBL close to the walls of an OC, whose recirculation regions are amplified by the interacting wakes of the obstacles. This condition seems coherent with the experimental data of Bezpalcova (2007, [12]) and with the numerical results of Sec.4, of Tominaga et al. (2013, [19]) and of Atzori et al. (2022, [66]). A possible solution is proposed in the following by introducing a Smoothed Wall Function for  $\bar{\varepsilon}$ .

Under OCs, the WF for  $\bar{\varepsilon}$  is generalised as a weighted average between the local estimation of a WF formulated for the NSBL and the local value of  $\bar{\varepsilon}$  computed by SPH eps-BE that is already affected by the WFs of the previous steps:

$$\bar{\varepsilon}_{WF} = \begin{cases} (1 - c_{OC}) \bar{\varepsilon}_{NSBL} + c_{OC} \bar{\varepsilon}_{BE}, & \frac{11\bar{\nu}}{u_*} < r \leq l_{WF} \\ 0, & 0 < r \leq \frac{11\bar{\nu}}{u_*} \end{cases} \quad (3.1)$$

where  $u_*$  (m/s) is the friction velocity,  $r$  (m) is the distance from the wall and  $l_{WF}$  (m) is the distance of the farther edge of the WF scope region from the same wall. In the absence of an OC, the WF is standard and its scope depends on the NSBL depth  $\delta_{NSBL, \bar{\varepsilon}}$  (m) as in Amicarelli (2024, [32]):

$$\bar{\varepsilon}_{NSBL} = \frac{u_*^3}{k_v r}, \quad \frac{11\bar{\nu}}{u_*} < r \leq \delta_{NSBL, \bar{\varepsilon}} \quad (3.2)$$

The weights in (3.1) are ruled by the canopy smoothing function  $c_{OC}$ , which has a power-law closed-form expression depending on the non-dimensional distance from the wall  $r/\Delta x$ , where  $\Delta x$  (m) is the size of the computational particles:

$$c_{OC} = \begin{cases} \left(\frac{r}{\Delta x}\right)^{\frac{1}{n_{OC}}}, & (OC), \quad l_{WF} = \Delta x, \quad n_{OC} \in \mathbb{N}_0, \quad n_{OC} (\text{default}) = 1000 \\ 0, & .not.(OC), \quad l_{WF} = \delta_{NSBL, \bar{\varepsilon}} \end{cases} \quad (3.3)$$

For this reason, the  $\bar{\varepsilon}$  WF (3.1)-(3.3) is interpreted as a Smoothed Wall Function. The canopy smoothing function  $c_{OC}$  plays a key role in an OC, whose recirculation regions drastically erode the NSBL, and is particularly relevant if the Pollutant Source (PS) is located close to a canopy wall. The formulation for  $c_{OC}$  is derived hereafter. First, one imposes a linear dependence of  $c_{OC}$  on  $r$ :

$$c_{OC} = c_1 r + c_2 \quad (3.4)$$

where  $c_1$  (m<sup>-1</sup>) and  $c_2$  are temporary constants. The Boundary Conditions (BCs) are set to have  $c_{OC}=1$  at the edge of the WF region in OCs ( $r=\Delta x$ ), where the local contribution of the NSBL WF (3.2) is null, and to reproduce  $c_{OC}=0$  at the edge of the NSBL ( $r=\delta_{NSBL, \bar{\varepsilon}}$ ), where (3.2) is recovered:

$$\begin{cases} c_{OC}|_{r=\Delta x} = 1 \\ c_{OC}|_{r=\delta_{NSBL, \bar{\varepsilon}}} = 0 \end{cases} \Rightarrow \begin{cases} c_1 = \frac{1}{\Delta x - \delta_{NSBL, \bar{\varepsilon}}} \\ c_2 = -\frac{\delta_{NSBL, \bar{\varepsilon}}}{\Delta x - \delta_{NSBL, \bar{\varepsilon}}} \end{cases} \Rightarrow c_{OC} = \frac{r - \delta_{NSBL, \bar{\varepsilon}}}{\Delta x - \delta_{NSBL, \bar{\varepsilon}}} \approx \frac{r}{\Delta x}, \quad \delta_{NSBL, \bar{\varepsilon}} \ll \min\{r, \Delta x\} \quad (3.5)$$

From numerical evidence (Sec.4), the linear model (3.5) provides results too close to the NSBL WF (3.2). The following power law is thus introduced to improve the linear model by varying  $c_{OC}$  within the WF region more smoothly:

$$c_{OC} = c_3 \left( \frac{r - \delta_{NSBL, \bar{\varepsilon}}}{\Delta x - \delta_{NSBL, \bar{\varepsilon}}} \right)^{\frac{1}{n_{OC}}} + c_4, \quad n_{OC} \neq 0 \quad (3.6)$$

where  $c_3$  (m<sup>-1</sup>) and  $c_4$  are temporary constants. After imposing the same BCs as (3.5), the formulation of (3.3) is obtained:

$$\begin{cases} c_{OC}|_{r=\Delta x} = 1 \\ c_{OC}|_{r=\delta_{NSBL, \bar{\varepsilon}}} = 0 \end{cases} \Rightarrow \begin{cases} c_3 = 1 \\ c_4 = 0 \end{cases} \Rightarrow c_{OC} = \left( \frac{r - \delta_{NSBL, \bar{\varepsilon}}}{\Delta x - \delta_{NSBL, \bar{\varepsilon}}} \right)^{\frac{1}{n_{OC}}} \approx \left( \frac{r}{\Delta x} \right)^{\frac{1}{n_{OC}}}, \quad \delta_{NSBL, \bar{\varepsilon}} \ll \min\{r, \Delta x\} \quad (3.7)$$

The negligibility of  $\delta_{NSBL}$  close to the bottom walls of an OC is experimentally verified in Bezpalcova (2007, [12]) from the vertical profiles of the friction velocity, both in the narrow and the wide street canyons of the MUST wind-tunnel experiment, which is numerically reproduced in the current study (Sec.4). As shown in Atzori et al. (2022, [66]), the experimental profiles of the friction velocity in the

cavity region of a flow past a wall-mounted cubic obstacle (Martinuzzi & Tropea, 1993, [67]) denote a very thin NSBL: their measurements seem to hold a finer spatial resolution than Bezpalcova (2007, [12]) due to a simpler domain; the absence of interacting wakes might limit the NSBL erosion in the recirculation regions. Unlike  $u^*$ , the experimental profile of the mean TKE can appreciably vary with the wall distance in the NSBL.

The selection criterion for the canopy coefficient  $n_{OC}$  in (3.7) detects the minimum  $n_{OC}$  value that prevents from representing the NSBL in the OC of a RANS simulation. This is verified by monitoring a representative vertical profile of the turbulent viscosity  $\nu_T$  ( $\text{m}^2/\text{s}$ ): it should be either nonlinear or its extrapolation to the wall from the closest particle value should be appreciable with respect to the top-canopy value, where  $H_b$  (m) is the local obstacle/building height:

$$n_{OC} = \min \{n_{OC,j}\} : \left[ \lim_{r \rightarrow 0} \nu_T|_r > o(\nu_T|_{r=H_b}) \right] \cup \left[ \frac{\partial \nu_T}{\partial r} \Big|_{r < \frac{H_b}{2}} \neq \text{const.} \right] \quad (3.8)$$

As the influence of  $n_{OC}$  on the  $\bar{\varepsilon}$  WF is relatively small, due to the time accumulation of  $\bar{\varepsilon}_{NSBL}$  contributions in (3.1), one can iteratively vary  $n_{OC}$  by an order of magnitude until the above criterion is satisfied, where the subscript ‘ $j$ ’ indicates a generic iteration. If this convergence procedure is repeated for each test case and each  $\Delta x$ , then the WF formulation is systematically consistent since the contribution of the OC becomes non-influential with  $\Delta x \rightarrow 0$ . Otherwise, the value  $n_{OC}=1000$ , obtained from the application of (3.8) to the MUST wind-tunnel test case (Sec.4) can be set as a default value for the sake of simplicity. Instead of using the physical criterion (3.8) adopted by the code of Sec.2, one might also calibrate  $n_{OC}$  to minimise the errors with respect to the available measurements, within the scope of data assimilation or back-casting analyses. The limit behaviours  $n_{OC} \rightarrow \infty$  and  $n_{OC}=0$  represent the absence of WF ( $c_{OC}=1$ ) and the NSBL WF ( $c_{OC}=0$ ), respectively.

The region of elastic impingements close to a fixed wall ( $r \leq 0.25\Delta x$ , Sec.2) contains a minority of all the nodes in the first particle layer close to the wall. At the outer edge of this region, one has  $c_{OC}$  ( $r=0.25\Delta x$ )=0.9986 and  $n_{OC}=1000$ . This means that, for the majority of those particles interested by the  $\bar{\varepsilon}$  WF in the presence of an OC, the following condition approximately applies to (3.1):

$$0 \leq (1 - c_{OC}) \bar{\varepsilon}_{NSBL} \leq \frac{\bar{\varepsilon}_{NSBL}}{1000} \quad (3.9)$$

This apparently small contribution of (3.2) to (3.1), seems sufficient to impose the proper BC, due to the time accumulation of  $\bar{\varepsilon}_{NSBL}$ , and not to disturb the representation of the Neutral cOre/Outer Boundary Layer (NOBL) near the canopy walls. Except for overrefined RANS simulations or specific test cases where the assumption of a negligible NSBL depth is misleading — the latter explicitly requires to set  $c_{OC}=0$  —, the two following extreme conditions should be avoided:  $c_{OC}=0$ ;  $c_{OC}=1$ . In the first case, the NSBL is reproduced in the first layer of particles close to the canopy walls:  $\bar{\varepsilon}$  is locally overestimated,  $\nu_T$  and the coefficient of turbulent diffusion  $K_T$  ( $\text{m}^2/\text{s}$ ) are under predicted with excessive under-diffusion of the pollutant mass from Ground-Level PSs. Under  $c_{OC}=1$ ,  $\bar{\varepsilon}$  BC is not properly imposed in a RANS simulation. NSBL is very thin and yet crucial for wall BCs. Since NSBL is numerically missing,  $\bar{q}$  grows overestimated without enough dissipation, whereas  $K_T$  determines excessive dispersion especially for those pollutants released by Ground-Level PSs.

The OC can be also represented by a complex boundary of a single obstacle, if the flow between its roughness elements has to be solved and these elements are configured as non-isolated obstacles. The combined effect of  $\bar{\varepsilon}$  WF and the associated SPH TKE-BE adaptation (Sec.2) is responsible for nonlinear effects. For example, the Smoothed Wall Function lowers  $\bar{\varepsilon}$  levels in the first layer of particles close to the canopy walls, with respect to the NSBL WF, but  $\bar{\varepsilon}$  might locally increase at larger distances within the SPH neighbourhood of the walls because the condition ( $P_q = \bar{\varepsilon}$ ) is not imposed out of the WF region, where  $P_q$  ( $\text{m}^2/\text{s}^3$ ) is the production rate of the mean TKE. Under OCs, the scope of the mean-velocity WF is not reduced in favour of free-slip conditions because the latter

would provide worse results in terms of accuracy, anti-penetration strategy and assessment of  $P_q$ . All the other attributes of the WFs are unchanged with respect to Amicarelli (2024, [32]).

#### 4. AIR QUALITY IN AND AROUND THE URBAN CANOPY OF THE WIND-TUNNEL MUST EXPERIMENT

##### 4.1. Numerical setting of the test case

The code of Sec.2 is validated on the wind-tunnel MUST experiment (Sec.1), under the configuration with the yaw angle of the building matrix set to  $45^\circ$ , with respect to the undisturbed mean velocity (Bezpalcova, 2007, [12]). The test case is here associated with a full-scale OAQ application featuring the accidental release of hydrocarbons ( $C_2H_6$ ) from a gas station and downstream transport into and around an UC of 36 buildings.

The experimental values used for input pre-processing at the domain boundaries and for validation in the inner domain refer to the PhD thesis of Bezpalcova (2007, [12]) and the dataset on the wind-tunnel MUST test case, realised in the EWTl laboratory and published within the frame of the COST 732 action (Leitl et al., 2007, [68]). The present study refers to a dataset version and the thesis that were freely and publicly available online in 2008.

The Region of Interest (ROI, Figure 1) of the current study contains 30 buildings, labelled as ‘G5’ to ‘L9’ in the dataset, out of the 120 buildings of the whole experiment. It is the upstream 25% of the wind-tunnel MUST Urban Canopy. The domain has the same orientation as the experiment, unlike Leuzzi et al. (2012, [11]), where the codes for the airflow dynamic and for the pollutant dispersion needed to align the obstacle axes with the reference system. The current orientation increases the geometry accuracy (the building matrix is not exactly homogeneous and isotropic), minimises the elapsed time (due to the size of the SPH positioning grid), permits an easier management of the velocity vectors, is more coherent with the BCs of the experiment at the lateral walls and is more suitable for possible changes in the yaw angle via the turning table to represent the other configurations of the same benchmark. In any case, under complex geometries, it is not feasible to exactly set the initial particle distribution over a uniform and isotropic Cartesian grid, which would minimise the SPH truncation errors. According to the experimental dataset and Bezpalcova (2007, [12]), the sizes of the buildings are approximately  $L_b \times W_b \times H_b = 0.163 \times 0.032 \times 0.034 \text{ m}^3$  — where  $L_b$  (m),  $W_b$  (m) and  $H_b$  (m) are the building length, width and height —, except for the shortest and tallest obstacle, called ‘VIP car’ in the dataset. This is located in the downstream area of the ROI and features the following sizes  $0.50L_b \times W_b \times 1.38H_b$ . The average inter-obstacle distances along the principal directions of the building matrix are  $5.06H_b$  and  $3.09H_b$ , according to the dataset. They are associated with wide and narrow street canyons, respectively. Compared to the RANS LSM-FDM simulation of Leuzzi et al. (2012, [11]) of the same test case, which took the airflow velocity fields as input data, here there is an additional obstacle row upstream from the PS and 6 further buildings (‘E7’, ‘F 6–8’ and ‘I–J 4’) out of the ROI to reproduce BCs better.

Considering a generic complex canopy of obstacles or buildings with proper average ‘ $_{avg}$ ’ values of  $L_b$ ,  $W_b$  and  $H_b$ , the domain characteristic lengths are set as follows. The domain depth can be set as  $H_{dom} = 5H_{b,avg}$ , following Bezpalcova (2007, [12]) that cites VDI (2005, [69]). They deemed such an assumption sufficient to assess the  $\bar{C}$  field within the UC, no matter the top BC. In general, this is valid only if the domain is short enough. The validity of this assumption on the current test case was confirmed by Leuzzi et al. (2012, [11]). Following Amicarelli (2024, [40]) on the characteristic distances between an isolated obstacle and the edges of the numerical domain, a generalisation is here provided in case a matrix of d-type or k-type roughness elements (e.g. an UC) is explicitly simulated under a wake-interference regime or a skimming regime. The matrix of buildings is assumed as a unique permeable obstacle. The minimum distance between the physical inlet section and the most upstream vertex of any obstacle of the UC is set equal to  $k_1 4 \max \{ H_{b,avg}; 0.5W_{b,avg} \}$ . The minimum distance between the outlet section and the most downstream vertex of any obstacle is set equal to



$10Ak_2$ , being  $A \equiv \max \{H_{b,avg}; 0.5L_{b,eq,avg}; 0.5W_{b,eq,avg}\} \cdot W_{b,eq}$  (m) and  $L_{b,eq}$  (m) are the equivalent width and length of a building:

$$W_{b,eq} = W_b \cos(\alpha_{yaw,b}) + L_b \sin(\alpha_{yaw,b}), \quad L_{b,eq} = L_b \cos(\alpha_{yaw,b}) + W_b \sin(\alpha_{yaw,b}) \quad (4.1)$$

where  $\alpha_{yaw}$  is the yaw angle of a generic building, here equal to  $\text{ca.}45^\circ$ . The minimum distance between each of the lateral symmetry planes of the domain and any obstacle vertex is expressed as  $k_3 9 \max \{0.5L_{b,eq,avg}; 0.5W_{b,eq,avg}\}$ . The regime of isolated obstacles is represented by  $k_1=k_2=k_3=1.00$ , which can be used as first-attempt values. Contrarily to  $H_b$ ,  $L_{b,eq}$  and  $W_{b,eq}$  are halved in the formulae above: they are built on the numerical results of Amicarelli (2024, [32]), where the bottom wall of the wall-mounted cube acts as a symmetry plane. After numerical tests, considering that the horizontal projections of all the monitoring points lie within the building matrix or upstream, and that the wake-interference regime occurs not far from the regime of isolated obstacles, the following values are obtained to minimise the elapsed time of the current test case:  $k_1=1.00$ ;  $k_2=0.10$ ;  $k_3=0.50$ . Thus, the domain sizes are  $L_{dom} \times W_{dom} \times H_{dom} = 14.3L_{b,eq,avg} \times 17.0W_{b,eq,avg} \times 5.00H_{b,avg}$ , net of the inlet region.

The reference mean speed is defined as  $U_{ref}=5$  m/s, whereas the value 7.89 m/s refers to the height of the Prandtl tube in the experiment. This lies over the top of the numerical domain. The largest mean speed in the numerical inlet section is  $\bar{u}_{inlet,max}=5.49$  m/s. The inlet mean speed at  $z=H_{b,avg}$  is  $\bar{u}=4.158$  m/s. The friction velocity obtained from the application of the NSBL WF to the measurements of the undisturbed mean velocity is  $u^*=0.33$  m/s (Bezpalcova, 2007, [12]). This is close to  $u^*=0.35$  m/s, derived from the NSBL values of the undisturbed vertical profile of  $\overline{u'w'}$  (Bezpalcova, 2007, [12]). The inlet value of the roughness length is  $z_{0,inlet}=2.2 \times 10^{-4}$  m (Bezpalcova, 2007, [12]). This is only valid for the inlet conditions: the roughness elements set on the bottom wall of the upstream subdomain are absent on the turning table hosting the building matrix, as shown by the photographs in Bezpalcova (2007, [12]).

The experimental values of  $u^*$  (from the undisturbed mean-velocity profile) and  $z_{0,inlet}$  feed the NSBL WF of the mean velocity to reconstruct  $\bar{u}$  properly all over the numerical inlet section, as verified in (Bezpalcova, 2007, [12]). The experimental value of  $u^*$ , as obtained from the undisturbed profile of  $\overline{u'w'}$ , is used to reconstruct  $\bar{q}$  and  $\bar{\varepsilon}$  at inlet using their NSBL and NOBL profiles, as in Amicarelli (2024, [32]). The input value for the BL depth  $\delta_T=0.800$  m of the wind-tunnel MUST experiment is derived following a dedicated procedure in Amicarelli (2025, [33]) for neutral conditions. Alternatively, for those stable and unstable atmospheric conditions of the open-field MUST experiment, out of the current scope, the inlet section might be set after Ferrand et al. (2025, [70]). Considering the photographs in Bezpalcova (2007, [12]), the coating of the turning table and the buildings is assumed not to be rougher than the material used earlier by Martinuzzi & Tropea (1993, [67]), for which  $H_{sgr}=1.35 \times 10^{-5}$  m was derived (Amicarelli, 2024, [32]). The quantification above corresponds to the sub-grid median diameter  $d_{50}=2.70 \times 10^{-5}$  m, which is close to  $1 \times 10^{-5}$  m, representative of the roughness of ‘rubber smooth tubing’ (Hall, 2012, [71]). This is compatible with the cited photographs. The above-mentioned value of  $H_{sgr}$  is finally assumed in input. Air density is set to  $\rho=1.22$  kg/m<sup>3</sup>, which is representative of the experimental conditions (temperature  $T=15^\circ\text{C}$  and atmospheric pressure; Bezpalcova, 2007, [12]). The dynamic viscosity is  $\mu=1.80 \times 10^{-5}$  Pa·s.

The following information on the experimental PS refers to Bezpalcova (2007, [12]). The gaseous passive pollutant is ethane ( $\text{C}_2\text{H}_6$ ) emitted from a Canyon PS. This is located at the Ground Level and represented by a cylinder with an open jacket, a closed top, the depth  $H_{PS}=0.004$  m and the barycentre height  $z_{PS}=0.002$  m. The emission speed is  $U_{jet,PS}=0.022$  m/s. Due to the PS shape, the local emission velocity holds a cylindrical symmetry so that the average emission velocity, which might be added to the local mean velocity, is a null vector. It was experimentally verified that the jet volume flow rate at PS  $Q_{vol,PS} \in [2.8 \times 10^{-6}; 2.8 \times 10^{-5}]$  m<sup>3</sup>/s was non-influential on the non-dimensional mean concentration. As a result, no explicit modelling of jet intrusion is mandatory. The experimental

dataset integrates the information on the PS as follows. The PS diameter is  $D_{PS}=0.010$  m. The horizontal coordinates of the PS barycentre are  $x_{PS}=0.3378$  m and  $y_{PS}=-0.1845$  m, after translation of the reference system to set the most upstream building vertex on the bottom wall as the origin of the reference system. The ‘typical values’ mentioned in the dataset for the flow rate at PS belong to the range  $Q_{vol,PS} \in [2.8 \times 10^{-6}; 3.9 \times 10^{-6}]$  m<sup>3</sup>/s. The PS sizes of the experimental dataset are confirmed by the photographs in Bezpalcova (2007, [12]), despite a possible typo in the text of the latter.

The numerical PS is set according to the emission scheme of Sec.6. The pollutant mass flow rate at PS is approximated as  $Q_{PS} = \text{ca. } \rho_p Q_{vol,PS} = 3.53 \times 10^{-6}$  kg/s, where the subscript ‘ $p$ ’ indicates the pollutant. Its volume flow rate at source  $Q_{vol,PS} = 2.78 \times 10^{-6}$  m<sup>3</sup>/s is coherent with the dataset and Bezpalcova (2007, [12]). Considering air density and the molar masses of both air and ethane, it follows that  $\rho_p = 1.27$  kg/m<sup>3</sup>, which allows to neglect buoyancy effects. The coefficient of molecular diffusion  $D_M$  (C<sub>2</sub>H<sub>6</sub> in air,  $T=15^\circ\text{C}$ ) =  $1.54 \times 10^{-5}$  m<sup>2</sup>/s is coherent with  $D_M$  (C<sub>3</sub>H<sub>8</sub> in air,  $T=0^\circ\text{C}$ ) =  $1.2 \times 10^{-5}$  m<sup>2</sup>/s (Secchi, 2013, [72]) and  $D_M$  (CH<sub>4</sub> in air,  $T=0^\circ\text{C}$ ) =  $1.6 \times 10^{-5}$  m<sup>2</sup>/s (Secchi, 2013, [72]). The spatial resolution is defined by  $\Delta x = H_{b,avg}/5 = 0.0068$  m and  $h/\Delta x = 1.30$ , where  $h$  (m) is the characteristic length of the SPH kernel support. Time resolution is ruled by the optimisation factor  $F_A = 4.80$  (Amicarelli, 2024, [32]). The kernel gradient is computed on the fly to reduce the Total Virtual Memory  $TVM$ . Lateral and outlet BCs are much approximated to minimise the elapsed time and simplify geometries. The top and lateral symmetry planes feature free-slip conditions. The bottom wall and the building faces represent WF-driven no-slip conditions, included the Smoothed Wall Function of Sec.3. A truncation outlet section (Amicarelli, 2025, [33]) is extracted from the downstream half of the right symmetry plane, only from its lowest part at  $z \leq H_{b,avg}$ , and permits to avoid more complicated SPH schemes for multiple outlet sections (e.g. Ferrand et al., 2017, [73]). Simulations are executed under reduced pressure for barotropic neutrally buoyant micro-scale airflows (Amicarelli, 2024, [32]). Initial Conditions (ICs) are managed as in Amicarelli (2024, [32]). The experimental dataset also contains the AQ network of 256 monitoring points, 73 of which lied within the current ROI, which was the most polluted region of the experimental domain. Here the AQ network is also partitioned into 17 bins along the  $y$  axis in the range  $y \in [-0.950; 0.750]$  m, with the uniform bin size  $\Delta y_{mon} = 0.100$  m. They allow to track the horizontal profiles HPs of  $\bar{C}$  used for validation (Figure 1). A vertical profile passing through the PS is used for inter-comparisons on  $\bar{\varepsilon}$  and  $Re_T$  to apply the selection criterion of the Smoothed Wall Function (Sec.4.2). The definition of the non-dimensional mean concentration  $\pi_{\bar{C}} \equiv \frac{\bar{C} U_{ref} H_b^2}{Q_{PS}}$  is consistent with Bezpalcova (2007, [12]).

Its maximum value at PS is  $\pi_{\bar{C}} = \text{ca. } 56$ . The dataset reports the precision  $E$  of the experimental values  $E(\pi_{\bar{C}}) = 0.003$ . It is used for validation, following the procedure described by Franke et al. (2008, [74]). The following validation metrics are assessed: the Fractional Bias ( $FB$ ); the Normalised Mean Square Error ( $NMSE$ );  $FAC2$ ; the *hit rate*; the Geometric Mean ( $GM$ ) and Variance ( $GV$ ). Each correlation plot is synthesised by the square of the Pearson correlation coefficient  $R^2$  and the constants of the regression line  $y_{meas} = a_r x_{sim} + b_{rl}$ , where  $y_{meas}$  indicates the axis of the measured values and  $x_{sim}$  is the axis of the corresponding simulated values.

Stationary conditions are ruled by a criterion based on the SPH-TSE filter (Amicarelli, 2024, [32]), which is satisfied at  $t = t_f = 0.80$  s. At that simulated time,  $2.8 \times 10^6$  SPH particles are present in the numerical domain. The optimum values for the reference background pressure (e.g. Isik & He, 2021, [75])  $p_{ref} = 33.2$  Pa and the bulk modulus  $K_b = 1.328 \times 10^3$  Pa are returned by their convergence procedures (Amicarelli, 2024, [32]). The maximum frozen flight time  $t_{FLY,FRO,max,PS} = \text{ca. } 0.239$  s is obtained by a convergence procedure presented in Amicarelli (2025, [33]). The SPPS scheme (Amicarelli, 2025, [33]) has negligible effects on the reference simulation.

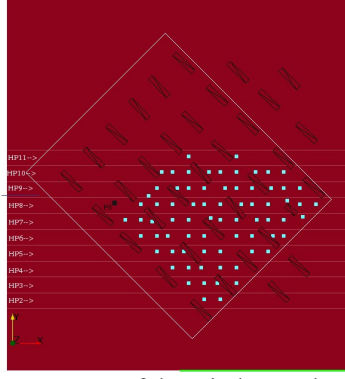


Figure 1. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Projection of boundaries and monitors on the horizontal: ROI (white-sided rectangle); buildings (black-sided rectangles); numerical PS (blue segment); physical PS and high/experimental-resolution vertical profiles (black square); AQ network (white dots); truncation outlet section (green segment). The bins of the horizontal profiles (HPs) are delimited by white segments in semi-transparency. The inlet undisturbed flow is aligned with x axis.

#### 4.2. Assessment of the canopy coefficient of the Smoothed Wall Function

The selection criterion for the Smoothed WF (Sec.3) applies to the current test case to assess the canopy coefficient  $n_{OC}$  (Figure 2). The simulations with  $n_{OC}=1$ ,  $n_{OC}=5$  and  $n_{OC}=100$  reproduce a NSBL along the portion of the vertical profile (Figure 1) lying in the Urban Canopy UC, in terms of  $Re_T = ca. 5 \nu_T / \nu$  (Figure 2, left panel). The near-wall behaviour is linear and the value extrapolated at the wall is null, as in the simulation with the NSBL WF (i.e.  $n_{OC}=0$ ). Despite appreciable differences are detected in terms of  $\bar{\varepsilon}$  (Figure 2, right panel), this quantity only influences the mean concentration via  $\nu_T$  and  $T_L$ . Reproducing a NSBL within the whole UC mismatches the experimental observations of Bezpalcova (2007, [12]) both in the narrow and the wide street canyons under the wake-interference regime. On the other end of the spectrum, a simulation with no WF ( $n_{OC} \rightarrow \infty$ ) does not match the proper BC showing an almost null gradient of  $\bar{\varepsilon}$  towards the wall, where the NSBL should be locally imposed. Following the selection criterion of Sec.3, with no tuning procedures,  $n_{OC}=1,000$  is returned as the reference value. It is the minimum value for which the NSBL is appreciably altered in the UC, as the extrapolated value of  $Re_T$  at the wall is not negligible with respect to the building top. The simulation with  $n_{OC}=10,000$  provides a more intense erosion of the NSBL but introduces a correction more invasive and arbitrary. The increase in  $n_{OC}$  determines a monotone growth of  $Re_T$  within the UC.

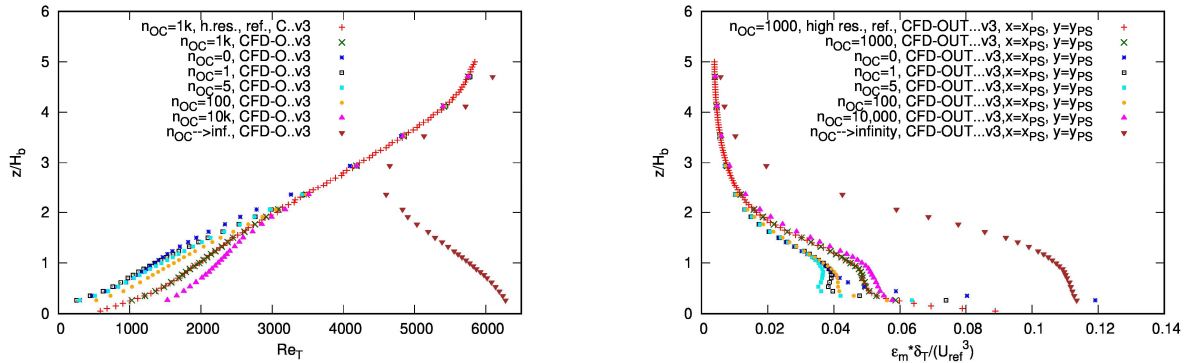


Figure 2. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Smoothed Wall Function: selection procedure for the exponent  $n_{OC}$ . Vertical profiles passing through the PS ( $t=t_f$ ). Left panel: turbulence Reynolds number  $Re_T$ ; right panel: mean-TKE dissipation rate  $\bar{\varepsilon}$ . Inter-comparisons between the reference simulation ( $n_{OC}=1,000$ ; high-resolution monitor +; experimental-resolution monitor  $\times$ ) and 6 auxiliary simulations with:  $n_{OC}=0$  (\*);  $n_{OC}=1$  ( $\square$ );  $n_{OC}=5$  ( $\blacksquare$ );  $n_{OC}=100$  ( $\circ$ );  $n_{OC}=10,000$  ( $\blacktriangle$ );  $n_{OC} \rightarrow \infty$  ( $\blacktriangledown$ ).

All the simulations converge for  $z \geq 2H_b$  both in terms of  $Re_T$  and  $\bar{\varepsilon}$ , except for the run without WF featuring residual overestimations. At the bottom of the vertical profile, the two extreme simulations provide close values of  $\bar{\varepsilon}$ , but highly different BCs, as determined by  $\bar{\varepsilon}$  gradient and the extrapolated value at the wall. The behaviour of the intermediate simulations is nonlinear: the minimum value of  $\bar{\varepsilon}$  is reproduced with  $n_{OC}=100$  at the lowest point of the experimental-resolution profile and with  $n_{OC}=5$  at the UC top. The high-resolution monitor passing through the PS permits a better representation of the profiles of the reference simulation than the experimental resolution. Compared to any auxiliary simulation with the NSBL WF (Figure 3, right panel), the Smoothed WF determines a systematic increase in the turbulence Reynolds number, whose peaks occur in the wake regions of the buildings (Figure 3, left panel). Such an increase shows cumulative effects with the downstream distance from the leading edge of the matrix of buildings.

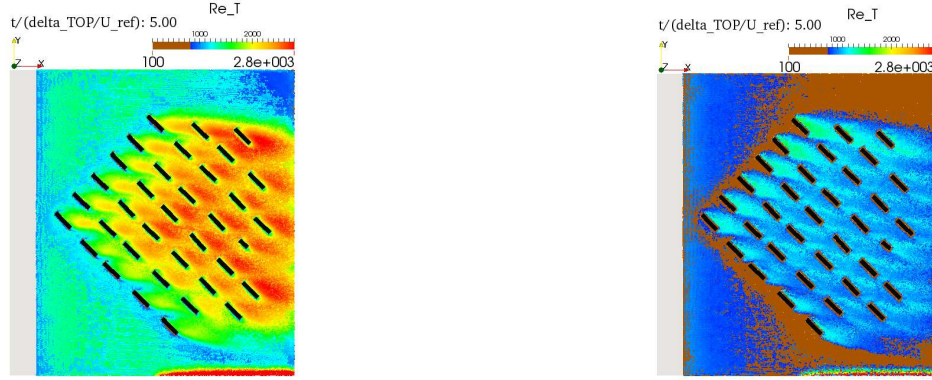


Figure 3. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Model inter-comparison: horizontal field of the turbulence Reynolds number (i.e. a non-dimensional turbulent viscosity) at  $z=0.1H_b$ . Left panel: reference simulation (CFD-OUT-sweePDF; Smoothed WF). Right panel: auxiliary simulation n.1 (CFD-OUT-sweePDF; NSBL WF).

#### 4.3. Validation on the mean concentration, inter-comparisons and full-scale application

The validation of the code of Sec.2 on the mean concentration of the wind-tunnel MUST experiment is documented in the present section, together with a sequence of model inter-comparisons and an associated full-scale application.

Figure 4 (top-left panel) shows the horizontal slice of the  $\pi_{\bar{c}}$  field at  $z=H_b/2$  as reconstructed over the points of the positioning grid via the SPH-TSE filter (Amicarelli, 2024, [32]), and the derived iso-lines. The centreline of the mean plume is partially channelised close to the PS and its direction rotates downstream, tending to align with the undisturbed mean velocity. This is due to the rotation of the mean flow direction with height and the progressive increase in the pollutant fluxes from above the UC. The concentration levels reduce of ca.50 times from the peak in the first polluted crossroad to the analogous position in the outlet section, where the plume laterally spreads over 4 wide street canyons. The vertical slice of the  $\pi_{\bar{c}}$  field passing through the PS is alternatively shown by either the grid points with the associated iso-lines (Figure 4, top-right panel), or the SPH computational nodes (Figure 4, bottom-right panel). The ballistic regime of turbulent transport is not appreciable, as the most polluted area downstream from the Ground-Level PS is associated with very small  $T_L$  values. The peak region at the UC top  $z=H_b$  occurs in the wide canyon of the PS ( $0.4 < \pi_{\bar{c}} < 1.0$ ). The height of the plume centreline apparently lifts with the downstream distance from the PS. This is due to the aforementioned rotation in the direction of the plume centreline. At the symmetry plane of the domain top,  $\pi_{\bar{c}}$  is smaller than 0.01 (i.e. 1/5,600 the PS value), and is also larger than 0.001 in the downstream half of the simulated domain, as confirmed by the iso-surface  $\pi_{\bar{c}}=0.001$  (Figure 4, bottom-left panel). The iso-surface  $\pi_{\bar{c}}=0.3$  seems fully contained within the UC. The iso-surface inter-comparison in

the 3D zoom clearly shows the rotation in the direction of the mean plume centreline downstream from the PS. The flow-canopy misalignment changes the direction of the plume centreline with the height and with the distance from the PS, and amplifies the forking of the most polluted portion of the plume within the building matrix. Both conditions affect  $P_q$  and  $\bar{C}$  gradients.

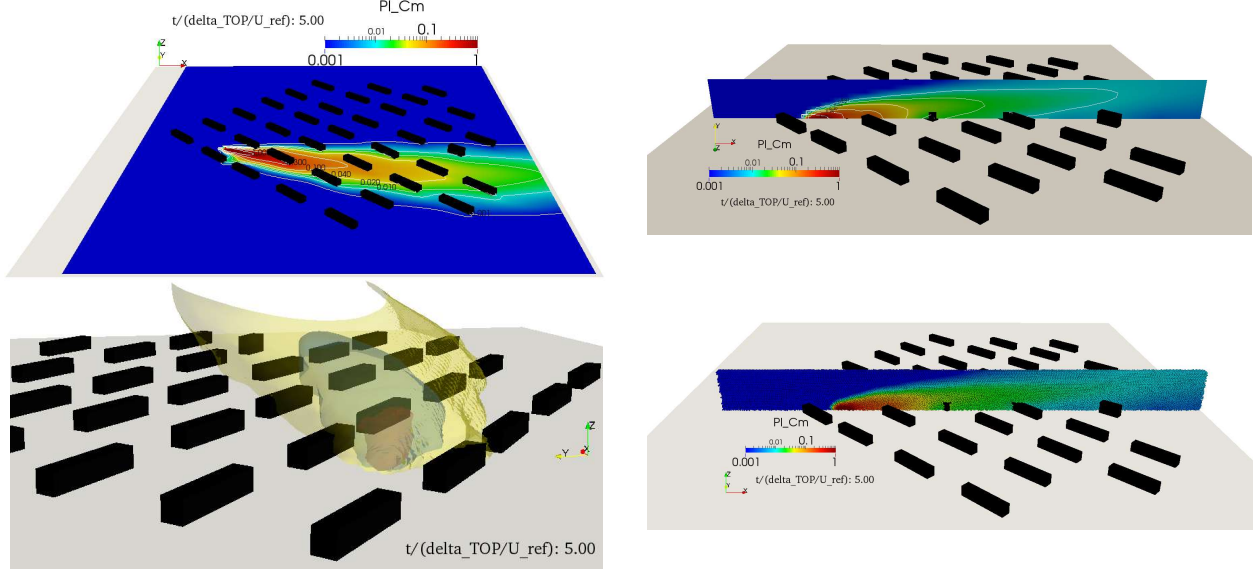


Figure 4. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Non-dimensional mean concentration  $\pi_{\bar{C}}$  at  $t=t_f$ . Top row: iso-lines reconstructed from the points of the positioning grid (top-left panel:

horizontal slice at  $z=0.5H_b$ ; top-right panel: vertical slice along  $y=y_{PS}$ ). Bottom-left panel: iso-surfaces at 0.300 (red), 0.040 (azure) and 0.001 (yellow). Bottom-right panel: vertical slice along  $y=y_{PS}$  featuring the SPH computational nodes.

Compared to the  $\pi_{\bar{C}}$  measurements of the AQ network at  $z=H_b/2$  (Bezpalcova, 2007, [12]) in the ROI (Figure 5, left panel), the reference simulation (Figure 5, centre panel) shows a positive graphical validation, especially upstream from the 2<sup>nd</sup> polluted narrow canyon. Downstream, a limited but systematic anticlockwise rotation of the plume penalises the reproduction of its right edge. With respect to the auxiliary simulation n.1, featuring a more traditional setting with a NSBL WF and a constant  $Sc_T$  (Figure 5, right panel), the combined use of the Smoothed WF (Sec.3) and of the  $Sc_T$ -TGS turbulence closure (Amicarelli, 2025, [33]) makes the reference simulation determine an increased pollutant diffusion and a limited extra rotation of the plume. Keeping the measurements as the actual field, the plume of the reference simulation is more accurate than the auxiliary simulation almost all over the domain and the overestimation in the extent of the most polluted area ( $\pi_{\bar{C}} > 0.3$ ) is drastically reduced. These improvements are due to considering the NSBL erosion close to walls under the wake-interference regime, the velocity autocorrelation and the dependence of  $Sc_T$  on  $Re_T$ . The field of the auxiliary simulation n.1 is similar to the RANS field of  $\pi_{\bar{C}}$  reported in Tominaga et al. (2013, [19]) where they used standard WFs and the constant  $Sc_T=0.90$ , and stretched a similar shortcoming of underestimated pollutant diffusion.

The reference simulation fairly reproduces the main horizontal profile of  $\pi_{\bar{C}}$  passing through the PS (HP8; Figure 6, top-left panel), despite a relevant local overestimation. In the following profile HP7 of the adjacent monitor bin with smaller  $y$  (Figure 6, top-right panel), the spatial evolution of  $\pi_{\bar{C}}$  is satisfactorily simulated, despite a limited but systematic overestimation due to a residual inhibition of the pollutant dispersion at the ground. In the following two profiles HP6 (Figure 6, bottom-left panel) and HP5 (Figure 6, bottom-right panel), the  $\pi_{\bar{C}}$  spatial evolutions and magnitudes are correctly simulated, whereas the rotation of the mean plume becomes appreciable (HP6) to relevant (HP5)



along the right edge of the plume. The combined use of the Smoothed WF (Sec.3) and of the  $Sc_T$ -TGS turbulence closure (Amicarelli, 2025, [33]) in the reference simulation permits to reproduce the left portion of the mean plume fairly, where the auxiliary simulation with SBL WF and constant  $Sc_T$  drastically underestimates the concentration levels. With this state-of-the-art setting, the simulated  $\pi_{\bar{C}}$  drops below its experimental precision, set as the lower bound of the y-axis in the plots, at the most points of the main horizontal profile HP8 (Figure 6, top-left panel).

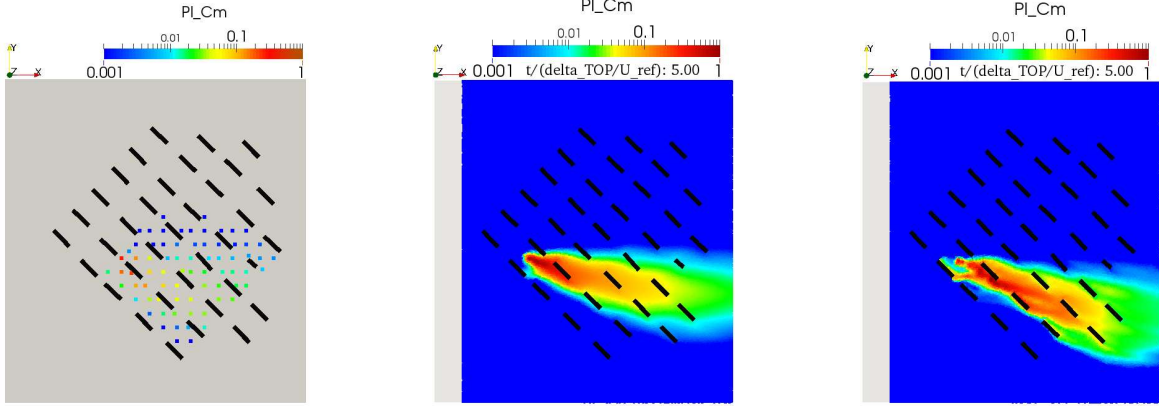


Figure 5. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Graphical validation and inter-comparison for the field of the non-dimensional mean concentration  $\pi_{\bar{C}}$  (horizontal slice at  $z=H_b/2$ ): measurements (Bezpalcova, 2007, [12], left panel); reference simulation (CFD-OUT-sweePDF; Smoothed WF and  $Sc_T$ -TGS; centre panel); auxiliary simulation n.1 (CFD-OUT-sweePDF; NSBL WF and  $Sc_T$ -const; right panel).

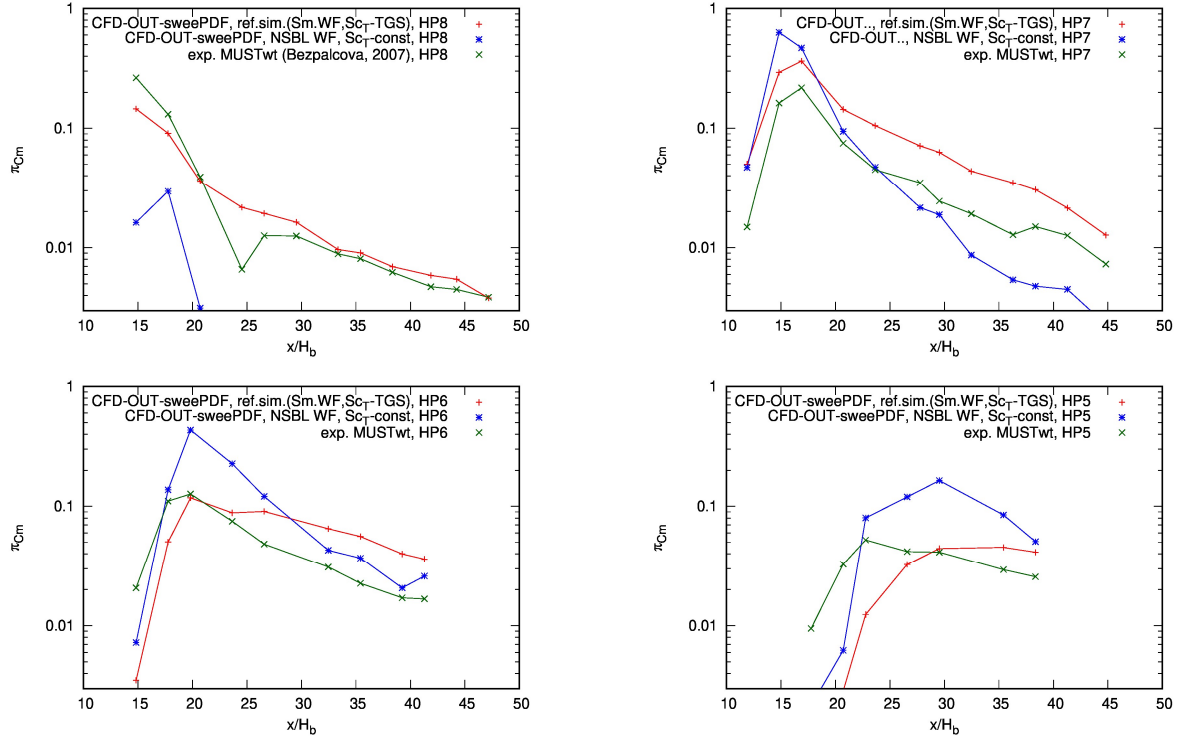


Figure 6. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Validation and inter-comparison for the non-dimensional mean concentration  $\pi_{\bar{C}}$ : horizontal profiles of the AQ network ( $z/H_b=0.5$ ) within the ROI with more than 3 measured points above the experimental precision. Reference simulation (CFD-OUT-sweePDF; Smoothed WF,  $Sc_T$ -TGS, +); auxiliary simulation n.1 (CFD-OUT-sweePDF; NSBL WF,  $Sc_T$ -const., \*); measurements (Bezpalcova, 2007, x). Profiles: HP8 (top-left panel); HP7 (top-right panel); HP6 (bottom-left panel); HP5 (bottom-right panel).

Further, the reference simulation enhances the spatial evolution of all the profiles and relevantly reduces the peak overestimations of the auxiliary simulation thanks to an increased diffusion at the ground and a more precise assessment of  $Sc_T$  (Figure 6, all the panels). As a secondary shortcoming, the excess in the plume rotation is more visible in the profiles HP5-6 of the reference simulation. The guideline bounds on  $FB$ ,  $NMSE$ ,  $FAC2$ ,  $GM$  and  $GV$  for the mean concentration (Table 4.1) are defined in Chang & Hanna (2004, [1]), whereas the analogous minimum threshold for the *hit rate* is set by VDI (2005, [69]). The latter appears as a clear anomaly because it is larger than the homologous  $FAC2$  value and is uncommonly satisfied by state-of-the-art models. In this context, the reference simulation meets all the state-of-the-art criteria for the validation metrics (considering the precision of the  $GV$  threshold), except for the *hit rate*, which is however larger than the corresponding  $FAC2$  threshold, as explained above.

| Dataset                                                                 | $FB$     | $NMSE$  | $FAC2$ (%) | <i>hit rate</i> (%) | $GM$    | $GV$    |
|-------------------------------------------------------------------------|----------|---------|------------|---------------------|---------|---------|
| Reference simulation<br>(CFD-OUT-sweepPDF;<br>Smoothed WF, $Sc_T$ -TGS) | -0.17    | 1.25    | 67         | 51                  | 0.99    | 1.70    |
| State-of-the-art upper bound                                            | 0.3±0.1  | 4.0±0.1 | 100        | 100                 | 1.3±0.1 | 1.6±0.1 |
| State-of-the-art lower bound                                            | -0.3±0.1 | 0.0     | 50±1       | 66±1                | 0.7±0.1 | 1.0     |

Table 4.1. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Validation metrics for the non-dimensional mean concentration  $\pi_{\bar{c}}$  of the AQ network ( $z/H_b=0.5$ ) within the ROI, by comparison with the measurements of Bezpalcova (2007, [12]). Data: reference simulation (CFD-OUT-sweepPDF; Smoothed WF and  $Sc_T$ -TGS); state-of-the-art bounds of Chang & Hanna (2004, [1]) for the Fractional Bias ( $FB$ ), the Normalised Mean Square Error ( $NMSE$ ),  $FAC2$ , the Geometric Mean ( $GM$ ) and Variance ( $GV$ ), and of VDI (2005, [69]) for the *hit rate*.

The introduction of the Smoothed WF and the use of  $Sc_T$ -TGS permit to improve all the validation metrics relevantly (Table 4.2) and to make  $FB$ ,  $NMSE$  and  $GV$  satisfy their state-of-the-art criteria.  $FB$  magnitude and  $NMSE$  are reduced of 65% and 78%. Considering two additional auxiliary simulations with either NSBL WF or  $Sc_T$ -TGS, it is demonstrated that most of the improvement (ca.80% for both  $FB$  and  $NMSE$ ) is due to the Smoothed WF, and that the remaining contribution (of ca.20%) is given by  $Sc_T$ -TGS, despite the small  $T_L$  values in the near field of  $\pi_{\bar{c}}$ . The absence of WF (auxiliary simulation n.4) makes the code drastically overestimate the pollutant diffusion.

| Dataset                                                                            | $FB$  | $NMSE$ | $FAC2$ (%) | <i>hit rate</i> (%) | $GM$ | $GV$ |
|------------------------------------------------------------------------------------|-------|--------|------------|---------------------|------|------|
| Reference simulation<br>(Smoothed WF, $Sc_T$ -TGS)                                 | -0.17 | 1.25   | 67         | 51                  | 0.99 | 1.70 |
| Auxiliary simulation n.1<br>(NSBL WF and $Sc_T$ -const)                            | -0.49 | 5.69   | 52         | 42                  | 1.14 | 2.14 |
| Auxiliary simulation n.2<br>(NSBL WF, $Sc_T$ -TGS)                                 | -0.42 | 4.51   | 59         | 49                  | 1.08 | 1.88 |
| Auxiliary simulation n.3<br>(Smoothed WF, $Sc_T$ -const)                           | -0.22 | 1.42   | 68         | 51                  | 1.00 | 1.82 |
| Auxiliary simulation n.4<br>( $n_{OC} \rightarrow \infty$ i.e. no WF, $Sc_T$ -TGS) | 0.48  | 3.44   | 29         | 16                  | 0.93 | 4.22 |

Table 4.2. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Validation metrics for the non-dimensional mean concentration  $\pi_{\bar{c}}$  of the AQ network ( $z/H_b=0.5$ ) within the ROI, by comparison with the measurements of Bezpalcova (2007, [12]): model inter-comparisons. Numerical data: reference simulation (CFD-OUT-sweepPDF; Smoothed WF and  $Sc_T$ -TGS); auxiliary simulations (CFD-OUT-sweepPDF; NSBL WF and/or  $Sc_T$ -const).  $FB$ : Fractional Bias;  $NMSE$ : Normalised Mean Square Error;  $GM$ : Geometric Mean;  $GV$ : Geometric Variance.

Table 4.3 compares the simulations of the current study with the 6 research/commercial CFD codes used in Tominaga et al. (2013, [19]). In both studies specific attention is paid not to introduce any tuning procedure in the code settings, at any level, even if commonly used or admitted by the state-

of-the-art procedures. The common validation metrics for  $\pi_{\bar{c}}$  are the *hit rate* and *FAC2*. The reference simulation obtains an above-average overall rank among the RANS codes, while it is inferior to the LES models. The combined use of the Smoothed WF and  $Sc_T$ -TGS turbulence closure makes the code of Sec.2 improve both metrics (especially *FAC2*) and its average overall rank  $Rank_{avg}$  from the last (6/7) to the second quartile (2.5/7). Its ranks are penalised because the current study only considers the most challenging 25% of the experimental domain, around the PS and the leading edge of the matrix of buildings, where  $\pi_{\bar{c}}$  is by far more inhomogeneous than the omitted subdomain.

| Ranking                                                                                                            | <i>FAC2</i><br>(%)                         | <i>hit rate</i><br>(%)                     | <i>Rank<sub>avg</sub></i> | Nodes ( $t=t_f$ )                                  |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------------|---------------------------|----------------------------------------------------|
| Virtual rank of ref. sim. in Tominaga et al. (2013, [19])<br>vs. 6 RANS codes (vs. 6 RANS + 2 LES codes)           | 2 <sup>nd</sup> /7<br>(4 <sup>th</sup> /9) | 3 <sup>rd</sup> /7<br>(5 <sup>th</sup> /9) | 2.5/7<br>(4.5/9)          | 2.8×10 <sup>6</sup> vs.<br>1.1–7.0×10 <sup>6</sup> |
| Virtual rank of aux. sim. n.1 in Tominaga et al. (2013, [19])<br>vs. 6 RANS codes (vs. 6 RANS codes + 2 LES codes) | 7 <sup>th</sup> /7<br>(9 <sup>th</sup> /9) | 5 <sup>th</sup> /7<br>(7 <sup>th</sup> /9) | 6/7<br>(8/9)              | (as above)                                         |

Table 4.3. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Number of nodes and validation metrics for the non-dimensional mean concentration  $\pi_{\bar{c}}$  by comparison with the measurements of

Bezpalcova (2007, [12]): model inter-comparisons. Numerical data: reference simulation (CFD-OUT-sweePDF; Smoothed WF and  $Sc_T$ -TGS); auxiliary simulation n.1 (CFD-OUT-sweePDF; NSBL WF and  $Sc_T$ -const); research and commercial CFD codes used in Tominaga et al. (2013, [19]) represented by 6 RANS k-eps and 2 LES models (LES Dynamic SGS and ILES). The ranks of the current study are penalised by only reproducing the most challenging 25% of the experimental domain.  $Rank_{avg}$ : average of the single ranks.

The simulations of the current study are also compared with the 20 research/commercial models used in the COST action 732 (Franke et al., 2008, [74]) in Table 4.4. The reference simulation obtains an above-average overall rank, with top-level peaks in *FB*, *NMSE* and *GM*. The combined use of the Smoothed WF and  $Sc_T$ -TGS turbulence closure makes the code of Sec.2 improve all its ranks and almost halves its average overall rank from 15.2/21 to 8.3/21. Even in this inter-comparison, the ranks of the current study are penalised by simulating the most challenging 25% of the experimental domain. This condition is particularly relevant for *FAC2*, *hit rate*, *GM* and *GV*, due to the role of the precision threshold of  $\pi_{\bar{c}}$  given to the validation procedure of Franke et al. (2008, [74]) and the relative weight of the lowest concentrations in assessing some metrics (e.g. *GV*).

| Ranking                                                                                      | <i>FB</i>            | <i>NMSE</i>          | <i>FAC2</i> (%)      | <i>hit rate</i> (%)  | <i>GM</i>           | <i>GV</i>            | <i>Rank<sub>avg</sub></i> |
|----------------------------------------------------------------------------------------------|----------------------|----------------------|----------------------|----------------------|---------------------|----------------------|---------------------------|
| Virtual rank of ref. sim. in<br>COST action 732 as reported in<br>Franke et al. (2008, [74]) | 2 <sup>nd</sup> /21  | 1 <sup>st</sup> /21  | 15 <sup>th</sup> /21 | 14 <sup>th</sup> /21 | 2 <sup>nd</sup> /21 | 16 <sup>th</sup> /21 | 8.3/21                    |
| Virtual rank of aux. sim. n.1 in<br>COST action 732 as per Franke<br>et al. (2008, [74])     | 14 <sup>th</sup> /21 | 11 <sup>th</sup> /21 | 20 <sup>th</sup> /21 | 17 <sup>th</sup> /21 | 9 <sup>th</sup> /21 | 20 <sup>th</sup> /21 | 15.2/21                   |

Table 4.4. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Validation metrics for the non-dimensional mean concentration  $\pi_{\bar{c}}$  by comparison with the measurements of Bezpalcova (2007, [12]): model inter-comparisons. Numerical data: reference simulation (CFD-OUT-sweePDF; Smoothed WF and  $Sc_T$ -TGS); auxiliary simulation n.1 (CFD-OUT-sweePDF; NSBL WF and  $Sc_T$ -const); 20 research and commercial models of the COST action 732 (Franke et al., 2008, [74]). The ranks of the current study are penalised by only reproducing the most challenging 25% of the experimental domain. *FB*: Fractional Bias; *NMSE*: Normalised Mean Square Error; *GM*: Geometric Mean; *GV*: Geometric Variance;  $Rank_{avg}$ : average of the single ranks.

In Table 4.5 the reference simulation is compared to the results of Leuzzi et al. (2012, [11]), obtained by the RANS LSM-FDM code LAGFLUM. This was validated on the same ROI as the current study and was alternatively fed by the RANS CFD-FVM code MISKAM and a meteorological diagnostic pre-processor that used the measurements internal to the domain. The first coupling should be preferred for a model inter-comparison. The performance of the code of Sec.2 is systematically



inferior to the state-of-the-art LSM, as expected when comparing deterministic with stochastic CFD on the same test case, where possible, in terms of accuracy (Amicarelli, 2024, [32]). Nonetheless, the advantages of the present code vs. the LAGFLUM-MISKAM coupling are reported in Sec.5.

| Dataset                                | <i>FB</i> | <i>NMSE</i> | <i>FAC2</i> (%) | <i>hit rate</i> (%) | <i>GM</i> | <i>GV</i> | Nodes ( $t=t_f$ )                   |
|----------------------------------------|-----------|-------------|-----------------|---------------------|-----------|-----------|-------------------------------------|
| Reference simulation (CFD-OUT-sweePDF) | -0.17     | 1.25        | 67              | 51                  | 0.99      | 1.70      | $2.8 \times 10^6$                   |
| MISKAM + LAGFLUM (stoch.)              | 0.02      | 0.84        | 88              | 63                  | 1.12      | 1.24      | n.a. + $1.0 \times 10^8$            |
| Diagnostic airflow + LAGFLUM           | -0.04     | 0.35        | 89              | 73                  | 0.97      | 1.23      | $5.1 \times 10^4 + 1.0 \times 10^8$ |

Table 4.5. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Number of nodes and validation metrics for the non-dimensional mean concentration  $\pi_{\bar{C}}$  of the AQ network ( $z/H_b=0.5$ ) within the ROI, by comparison with the measurements of Bezpalcova (2007, [12]): model inter-comparisons. Numerical data: reference simulation (CFD-OUT-sweePDF); LAGFLUM (Leuzzi et al., 2012, [11]) fed by CFD-FVM code MISKAM for the airflow (Balczo et al., 2009, [76]; Eichhorn & Balczo, 2008, [77]); LAGFLUM (Leuzzi et al., 2012, [11]) fed by measurement pre-processing for the airflow. *FB*: Fractional Bias; *NMSE*: Normalised Mean Square Error; *GM*: Geometric Mean; *GV*: Geometric Variance.

The correlation plot for  $\pi_{\bar{C}}$  of the reference simulation shows a good agreement with the measurements, accounting for the complexity of the test case (Figure 7). There is a limited but systematic overestimation at any concentration level, but the largest errors refer to a minority of under-predictions along the right edge of the mean plume. Many assessments slightly miss the *FAC2* satisfaction area. Compared to the auxiliary simulation n.1, featuring the NSBL WF and  $Sc_T$ -const, the combined use of the Smoothed WF and the  $Sc_T$ -TGS turbulence closure systematically reduces the errors, independently of their sign and magnitude, and permits to extend the range of *FAC2* satisfaction from  $[0.03, 0.05]$  to  $[0.01, \pi_{\bar{C}, \max, meas}]$ , as per the regression lines.

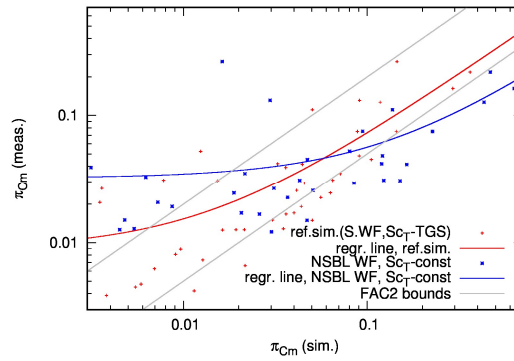


Figure 7. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Correlation plots for the non-dimensional concentration  $\pi_{\bar{C}}$  of the AQ network ( $z/H_b=0.5$ ) within the ROI, by comparison with the measurements of Bezpalcova (2007, [12]). Legend: reference simulation (CFD-OUT-sweePDF; Smoothed WF,  $Sc_T$ -TGS; +); auxiliary simulation n.1 (CFD-OUT-sweePDF; NSBL WF,  $Sc_T$ -const; \*). The plots focus on the couples of values above the experimental precision; the regression lines also consider the points of negligible values out of the visible window. *FAC2* bounds are shown.

Considering the coefficients of the regression lines of the correlation plots of Figure 7 and the square of the associated correlation coefficients  $R^2$  (Table 4.6), the reference simulation relevantly improves the correlation between simulated and measured values up to  $R^2=0.71$  and reduces the systematic overestimations of the auxiliary simulation n.1, representing a state-of-the-art configuration. In terms of these statistics, the improvements are only due to the use of the Smoothed WF, as demonstrated by the comparison to the auxiliary simulations n.2 and n.3. It is confirmed that the absence of WF (auxiliary simulation n.4) makes the code relevantly overestimate the pollutant diffusion.

| Simulation                                                                      | $R^2$ | $a_{rl}$ | $b_{rl}$ |
|---------------------------------------------------------------------------------|-------|----------|----------|
| Reference simulation (Smoothed WF, $Sc_T$ -TGS)                                 | 0.71  | 0.666    | 0.0056   |
| Auxiliary simulation n.1 (NSBL WF, $Sc_T$ -const)                               | 0.46  | 0.303    | 0.0136   |
| Auxiliary simulation n.2 (NSBL WF, $Sc_T$ -TGS)                                 | 0.49  | 0.351    | 0.0127   |
| Auxiliary simulation n.3 (Smoothed WF, $Sc_T$ -const)                           | 0.70  | 0.622    | 0.0061   |
| Auxiliary simulation n.4 ( $n_{OC} \rightarrow \infty$ i.e. no WF, $Sc_T$ -TGS) | 0.45  | 2.05     | -0.007   |

Table 4.6. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Coefficients of the regression lines  $y_{meas}=a_{rl}x_{sim}+b_{rl}$  and square of the correlation coefficient ( $R^2$ ) for the correlation plots of the non-dimensional mean concentration  $\pi_{\bar{C}}$  of the AQ network ( $z/H_b=0.5$ ) within the ROI. Data: reference simulation (CFD-OUT-sweepPDF; Smoothed WF,  $Sc_T$ -TGS); auxiliary simulations (CFD-OUT-sweepPDF, NSBL WF and/or  $Sc_T$ -const); measurements (Bezpalcova, 2007, [12]).

The simulated turbulent Schmidt number  $Sc_T$  at the PS level lies in the range  $0.3 \leq Sc_T \leq 0.6$  within the volume of the mean plume, except for a very small region around the PS (Figure 8, left panel). The comparison to the state-of-the-art value  $Sc_T=0.72$  (e.g. Violeau, 2009, [78]) of the  $Sc_T$ -const approach explains why the  $Sc_T$ -TGS turbulence closure here increases the diffusion of the mean concentration also near the PS. The small values of  $Sc_T$  in the near field are motivated by very small  $T_L$  values and increased  $\tilde{t}_{FP}$  values, both due to the proximity of the PS with the bottom wall (Figure 8, centre panel).

At the emission level, the  $\tilde{t}_{FP}/T_L$  ratio is larger than the unity all over the mean plume (except for a negligible area around the PS) so that the ballistic regime of pollutant transport is not detected. Thus, the improvements introduced by  $Sc_T$ -TGS are limited in the current test case, if compared to a possible configuration with an elevated PS. The main recirculation regions, where the weighted mean of the pollutant fly time cumulates, are marked by the local peaks in the  $\tilde{t}_{FP}/T_L$  ratio. This grows with the downstream distance from the PS and reaches a maximum of  $\tilde{t}_{FP}/T_L=164$  ( $z/H_b=0.1$ ). The regime of turbulent diffusion ( $\tilde{t}_{FP}/T_L > \text{c.a. } 20$ , i.e. from red to brown in Figure 8, centre panel) covers most of the mean plume area at the emission height. The  $\tilde{t}_{FP}/T_L$  ratios are partially damped in the wake regions where the peak value of the compensated velocity Lagrangian structure function  $\tilde{C}_0$  shows its maxima (Figure 8, right panel). The values of  $\tilde{C}_0$  lie within the range  $4.0 < \tilde{C}_0 < 5.2$  at the emission level. This quantity is here used to assess  $T_L$  and  $Sc_T$  (Amicarelli, 2025, [33]). In stochastic CFD,  $\tilde{C}_0$  is not computed, but considered as a numerical constant (equal to 2.1, Pope, 1985, [79]) of the turbulence closure of the BE for the momentum pdf. In other contexts,  $\tilde{C}_0$  is assumed equal to the Kolmogorov constant  $C_0=6.5$ , which should be the limit behaviour of  $\tilde{C}_0$  for  $Re_T \rightarrow \infty$ . Sawford et al. (2008, [80]) clarified the dependence of  $\tilde{C}_0$  on  $Re_T$  (Figure 8, right panel), providing a powerful closed-form solution for the direct assessment of  $\tilde{C}_0$ , used by the code of Sec.2.

The computational cost of the reference simulation is represented by the Total Virtual Memory  $TVM=22.1$  GB and the elapsed time  $t_{ela}=83\text{h}59\text{min}$  on 3.5 physical cores (with hyper-threading) of an i9-13900HX CPU processor. The very small number of cores is only due to an operative strategy at a very low cost. Towards the other edge of massively parallelised SPH runs, Cui et al. (2020, [81]) simulated the impingement and solidification of Super-cooled Large Droplets on aircraft surfaces with 1024-core runs.

The wind-tunnel MUST test case is here associated with a full-scale application concerning the accidental release of hydrocarbons ( $C_2H_6$ ) from a gas station with downstream transport into and around an Urban Canopy of 36 buildings. The 3D representation with progressive colouring of the simulated field of the mean concentration (Figure 7) highlights the most polluted volume of the

airflow after  $t=600s$  since the start of the continuous release. It also shows the incremental rotation of the centreline of the mean plume with the downstream distance from the PS.

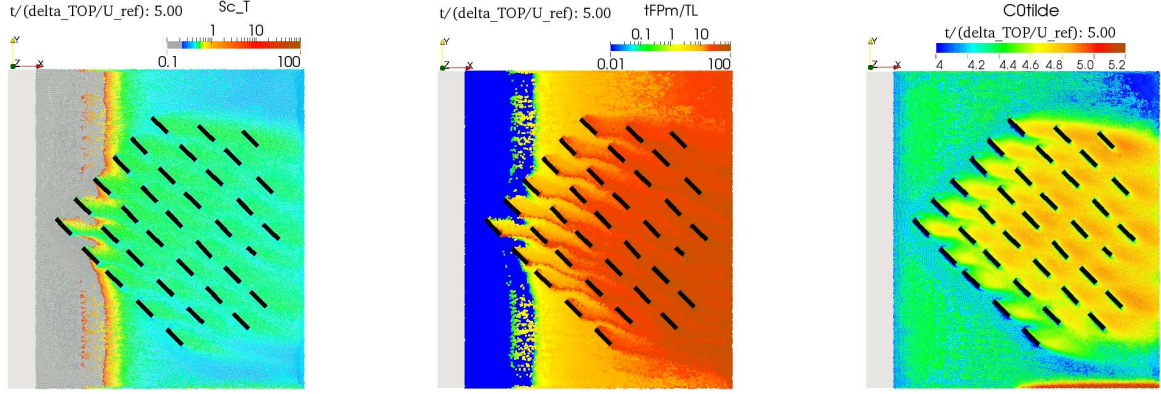


Figure 8. Air Quality in and around the Urban Canopy of the wind-tunnel MUST test case. Turbulent Schmidt number  $Sc_T$  (left panel) and  $\tilde{t}_{FP}/T_L$  ratio (centre panel): horizontal slices at  $z=0.1H_b$ . Peak value of the compensated velocity Lagrangian structure function  $\tilde{C}_0$ : horizontal slice at  $z=0.5H_b$  (right panel). Reference simulation (CFD-OUT-sweePDF; Smoothed WF,  $Sc_T$ -TGS).

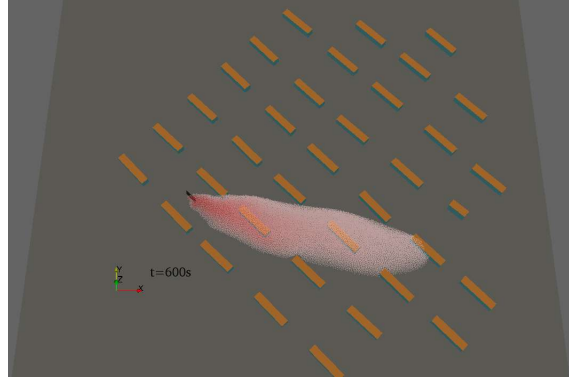


Figure 9. Full-scale application: accidental release of hydrocarbons ( $C_2H_6$ ) from a gas station and downstream transport into and around an Urban Canopy of 36 buildings (cyan: façades; brown: roofs; dark grey: roads). Mean concentration  $\bar{C}$  at  $t=600s$  goes from transparent white for null values to saturated red at the black PS (CFD-OUT-sweePDF v.3.0.0).

## 5. CONCLUSIONS

A Lagrangian RANS CFD-SPH code is validated on the Urban Canopy of the wind-tunnel MUST experiment (Bezpalcova, 2007, [12]) in terms of mean concentration  $\bar{C}$ , exploring a new Outdoor Air Quality application for the SPH numerical method. The flow-canopy misalignment in the current test case is a further challenge as it amplifies the forking of the most polluted portion of the plume within the building matrix and modifies the direction of the plume centreline with the height and with the distance from the Pollutant Source. The code, here improved by a Smoothed Wall Function, satisfies the thresholds of Chang & Hanna (2004, [1]) defined for five validation metrics to select state-of-the-art numerical models in assessing  $\bar{C}$ .

The current code shows an overall above-average performance after inter-comparisons with 28 deterministic research/commercial codes applied to the same test case (Tominaga et al., 2013, [19]; Franke et al., 2008). They represent current state-of-the-art models for OAQ in and around UCs. Further, the ranks of the present code are penalised because it only considers the most challenging 25% of the experimental domain, around the PS and the leading edge of the matrix of buildings, where  $\bar{C}$  is by far more inhomogeneous than the omitted subdomain.

The current code relies on the Kat&Lau-k-eps turbulence closure for the Momentum Equation and the  $Sc_T$ -TGS closure for the Balance Equation of  $\bar{C}$ . The improvement of k-eps models under specific

conditions is a recurrent practice in CFD modelling. RANS 2-equation 1st-order turbulence models are widely used for practical applications (Koukouvinis et al., 2016, [82]) due to their low computational cost and versatility; nonetheless, they need specific corrections under certain conditions (e.g. Kat&Lau-k-eps reduces the mean-TKE overestimations in deformation-dominated regions as around stagnation points) or let more complex turbulence approaches go ahead (e.g. 7-equation 2<sup>nd</sup>-order RANS models, LES models) to represent specific classes of phenomena (e.g. the onset of cavitation, Koukouvinis et al., 2016, [82]). Here Kat&Lau-k-eps is integrated by a Smoothed Wall Function for the pollutant transport into an Obstacle/Urban Canopy, as reported in the following. The systematic application of a standard WF for  $\bar{\varepsilon}$  to those nodes of a RANS simulation close to a wall seems incoherent with the intense erosion of the Neutral Surface Boundary Layer in an OC, whose recirculation regions are amplified by the interacting wakes of the obstacles. The consequent decrease in the coefficient of turbulent diffusion  $K_T$  seems to have drastic under-diffusion effects on pollutants emitted by Ground-Level PSs. This conclusion is supported by comparing the experimental data of Bezpalcova (2007, [12]) with the numerical results of Tominaga et al. (2013, [19]), Atzori et al. (2022, [66]) and the present study. Tominaga et al. (2013, [19]) validated 6 RANS and 2 LES CFD mesh-based models on the wind-tunnel MUST experiment. All their RANS simulations highlighted a systematic and relevant under-diffusion of the pollutant mass downstream from the Ground-Level Pollutant Source, with respect to the associated measurements and LES estimates. They used a standard NSBL WF and the constant turbulent Schmidt number  $Sc_T=0.9$ . An analogous behaviour is shown by the auxiliary simulation n.1 of the present study featuring a NSBL WF and the constant  $Sc_T=0.72$  ( $Sc_T$ -const approach). In both studies specific attention is paid not to introduce any tuning procedure in the code settings, even if commonly used or admitted by the state-of-the art procedures. A possible solution to the RANS issue above is the introduction of a Smoothed Wall Function for  $\bar{\varepsilon}$  under the wake-interference and the skimming regimes of OCs. The WF for  $\bar{\varepsilon}$  is generalised as a weighted average between the local estimation of a NSBL WF and the local value of  $\bar{\varepsilon}$  computed by its BE. The Smoothed WF permits a progressive switch from no WF for a node position on the outer edge of the WF scope region, whose depth is reduced to the node size under OCs, to the NSBL WF if the node position reaches the wall. The integral application of either of these two extreme conditions to all the nodes in the WF region seems detrimental because it completely neglects either the proper BC or the actual nature of the flow where the WF applies. The integral application of a standard NSBL WF determines systematic  $\bar{\varepsilon}$  excess at walls, associated with underestimations of  $K_T$  that relevantly inhibit the diffusion of pollutants emitted by a Ground-Level PS (auxiliary simulation n.1). Under the same conditions, the absence of WF makes the code assess very low levels of  $\bar{\varepsilon}$ , with  $K_T$  overestimations that drastically amplify pollutant diffusion (auxiliary simulation n.4). The adaptation of the mean-TKE BE to the  $\bar{\varepsilon}$  WF (Amicarelli, 2024, [32]) supports the solution above. The formulation of a WF for OCs, i.e. under the wake-interference regime and the skimming regime of non-isolated obstacles, seems a novelty in CFD modelling and might interest any numerical method and any RANS model no matter its turbulence closure.

$Sc_T$ -TGS (Amicarelli, 2025, [33]) is a closure to the mean-concentration Balance Equation on the turbulent Schmidt number  $Sc_T$ . It is coherent with Taylor's theory (Taylor, 1922, [44]), grid turbulence and shear flows, and reproduces both the effects of velocity autocorrelation and the dependence of  $Sc_T$  on the turbulence Reynolds number  $Re_T$ . Here  $Sc_T$ -TGS is first validated in the presence of an Obstacle/Urban Canopy, by comparison with the  $Sc_T$ -const state-of-the-art approach. The simulated  $Sc_T$  at the PS level lies in the range  $0.3 \leq Sc_T \leq 0.6$  within the volume of the mean plume, except for a very small region around the PS. The comparison to  $Sc_T=0.72$  of  $Sc_T$ -const explains the  $Sc_T$ -TGS beneficial increase in the diffusion of the mean concentration, also near the PS. The small  $Sc_T$  in the near field is due to very small values of the Lagrangian integral time scale of turbulence  $T_L$  and large values of the weighted mean of the pollutant flight time  $\bar{t}_{FP}$ , with a PS so close to the bottom wall. At the emission height, the  $\bar{t}_{FP}/T_L$  ratio is larger than the unity all over the mean plume (except

for a negligible area around the PS) so that the ballistic regime of pollutant transport is not detected. Thus, the improvements given by  $Sc_T$ -TGS to this study would be even larger with an Elevated PS. With respect to the auxiliary simulation n.1 mimicking the state-of-the-art with a NSBL WF and a constant  $Sc_T$ , the combined use of the Smoothed WF and the  $Sc_T$ -TGS turbulence closure makes the reference simulation of the current study determine increased pollutant diffusion and an anticlockwise rotation of the plume. Keeping the measurements as the actual field, the plume of the reference simulation is relevantly more accurate than the auxiliary simulation all over the domain, as confirmed by the validations, inter-comparisons and metrics here assessed. The overestimation in the extent of the most polluted area is drastically shrunk. The Fractional Bias magnitude and the Normalised Mean Square Error are reduced of 65% and 78%, respectively; 80% of both reductions is due to the Smoothed WF, 20% to  $Sc_T$ -TGS. In terms of correlation plots for the non-dimensional mean concentration  $\pi_{\bar{C}}$ , the domain of  $FAC2$  satisfaction enhances from  $[0.03, 0.05]$  to  $[0.01, \pi_{\bar{C}, \max, meas} = 0.264]$ , as per the regression lines, whereas the square of the correlation coefficient  $R^2$  increases from  $R^2=0.46$  to  $R^2=0.71$ . In the two model inter-comparisons with other 28 codes, the average ranks of the current code are approximately halved.

The performance of the code of Sec.2 is inferior to the state-of-the-art RANS LSM-FDM code LAGFLUM (Leuzzi et al., 2012, [11]), fed the RANS CFD-FVM results (Balczo et al., 2009, [76]; Eichhorn & Balczo, 2008, [77]) of the code MISKAM, as expected when comparing deterministic with stochastic CFD on the same test case, where possible, in terms of accuracy. Nonetheless, some advantages of the present model are highlighted hereafter. A sole code with a sole set of computational nodes is used. There is no mesh to prepare, not even for the airflow. No constraint applies to the shape of the buildings. No data assimilation procedure is adopted, even if accepted by the state of the art, either for the airflow dynamic or the pollutant transport. With respect to the LAGFLUM-MISKAM coupling, the code of Sec.2 also features general benefits besides the current inter-comparison, but it does not assess the concentration turbulent fluctuations, which is a major feature of LAGFLUM. The interfaces are directly estimated by the current code. The dynamic of solid bodies transported by fluid flows is simulated. The WC approach permits explicit time integration of independent PDEs. The mean advection terms of the BEs are freely and exactly represented thanks to a fully Lagrangian approach also for the airflow dynamic. The code is also versatile to full couplings of numerical schemes. The applications to Indoor and Duct Air Quality have also been investigated. The code potentially acts as a general-purpose CFD model besides meteorology and AQ applications, switches from RANS to ILES or Navier-Stokes' BEs, and treats fixed and mobile solid boundaries of any complex shape.

The current study also formulates and applies mathematical expressions to design the optimum characteristic lengths of the numerical domain in the presence of an OC. It is the distances of the inlet/outlet sections and the symmetry planes from the closest obstacles.

The peak value of the compensated velocity Lagrangian structure function  $\tilde{C}_0$  lies within the simulated range  $4.0 < \tilde{C}_0 < 5.2$  at the emission height.  $\tilde{C}_0$  is here used to assess  $T_L$  and  $Sc_T$ . In stochastic CFD (e.g. RANS-PDF; RANS-SPDF; Lagrangian Stochastic Models like Thomson, 1987, [13]),  $\tilde{C}_0$  is not computed, but considered as a numerical constant (equal to 2.1 in Pope, 1985, [79]) of the turbulence closure of the BE for the momentum pdf. In other contexts,  $\tilde{C}_0$  is assumed equal to the Kolmogorov constant  $C_0=6.5$ , which should be the limit behaviour of  $\tilde{C}_0$  for  $Re_T \rightarrow \infty$ . Sawford et al. (2008, [80]) clarified the dependence of  $\tilde{C}_0$  on  $Re_T$ , providing a powerful closed-form solution for the direct assessment of  $\tilde{C}_0$ , as used by the code of Sec.2. Sawford et al. (2008, [80]) confirmed that the practical advantages of the stochastic model of Sawford (1991, [83]) for acceleration, more accurate than Langevin equation but more expensive, might be given to stochastic models for velocity by replacing any  $C_0$  constant (either physical or numerical) with the direct assessment of  $\tilde{C}_0$ .

The test case of the present study is here associated with a full-scale OAQ application featuring the accidental release of hydrocarbons ( $C_2H_6$ ) from a gas station and downstream transport into and around an UC of 36 buildings. Besides the scope of the present paper, the accident dynamic triggering the emission of LNG from a transportation or storage tank might be also simulated by the CFD-SPH numerical method (e.g. Luo et al., 2020, [84]).

### Acknowledgements

Formal details on the code CFD-OUT-sweePDF v.3.0.0 (Amicarelli et al.) are reported at <https://www.researchgate.net/profile/Andrea-Amicarelli> and [github.com/AndreaAmicarelli-personal](https://github.com/AndreaAmicarelli-personal). Videos on the results of CFD-OUT-sweePDF are published at <https://www.youtube.com/@andrea.amicarelli>. The first preprint of the current study was published on 31 October 2025.

With exclusive regard to those source-code parts which were imported in CFD-OUT-sweePDF v.1.0.0 (Amicarelli et al.) from SPHERA v.10.0.0 (RSE SpA) — as mentioned twice in the last lines of Sec.2, with detailed references to the associated papers — the following notice applies: ‘SPHERA v.10.0.0 was realised by RSE SpA thanks to the funding ‘Fondo di Ricerca per il Sistema Elettrico’ within the frame of a Program Agreement between RSE SpA and the Italian Ministry of Economic Development (Ministero dello Sviluppo Economico)’.

The experimental values used for input pre-processing at the domain boundaries and for validation in the inner domain refer to the PhD thesis of Bezpalcova (2007, [12]) and the dataset on the wind-tunnel MUST test case, realised in the EWTl laboratory and published within the frame of the COST 732 action (Leitl et al., 2007, [68]). The present study refers to a dataset version and the thesis that were freely and publicly available online in 2008.

### Disclosure statement

The author reports there are no competing interests to declare.

## 6. APPENDIX. EMISSION SCHEME, THE NUMERICAL MEAN CONCENTRATION AT THE POLLUTANT SOURCE

The code of Sec.2 adopts a simpler emission scheme than mesh-based CFD codes (e.g. Zhang et al., 2024, [29]) to represent the PS, exploiting the Lagrangian attribute of the SPH method for the sake of efficiency. The emission model for the particle mean concentration at the PS represents a spatial reconstruction scheme which applies if the following conditions are met: the mean pollutant-mass flow rate at source  $\bar{Q}_{PS}$  (kg/s) is known; either the emission mean velocity is assumed uniform or the explicit modelling for jet intrusions in the atmosphere is not mandatory. The resulting model is an emission scheme under a uniform (specific mean) pollutant-mass flow rate. Unless otherwise stated, the PS is hereafter a point or a surface source, and lies within the physical inlet section or slightly downstream, as clarified in the following.

A physical PS features its input flow rate  $\bar{Q}_{PS}$  and area  $A_{PS}$  ( $m^2$ ), and assumes a uniform specific mean pollutant-mass flow rate at source  $\bar{\Theta}_{PS}$  (kg/s/ $m^2$ ). The resulting  $\bar{C}$  at PS varies with  $\bar{u}$  over the local coordinates  $\eta$  (m) and  $\zeta$  (m) within the plane of the PS. This is assumed normal to the direction of the spatially averaged mean velocity around the PS. In the discrete, the equivalent ‘ $_{eq}$ ’ numerical source adopts analogous conditions and represents the correct physical flow rate  $\bar{Q}_{PS}$  by construction:

$$\left\{ \begin{array}{l} \bar{Q}_{PS,eq} = \sum_{iPS=1}^{n_{pPS}} \bar{\Theta}_{PS,eq} \Delta \eta_{iPS} \Delta \zeta_{iPS} \approx \sum_{iPS=1}^{n_{pPS}} \bar{C}_{iPS} \left( -\underline{n}_{inlet} \cdot \bar{\underline{u}}_{iPS} \right) \Delta \eta_{iPS} \Delta \zeta_{iPS} \\ \bar{\Theta}_{PS,eq} \text{ uniform} \Rightarrow \bar{\Theta}_{PS,eq} = \frac{\bar{Q}_{PS,eq}}{A_{PS,eq}}, \quad A_{PS,eq} \equiv \sum_{iPS=1}^{n_{pPS}} \Delta \eta_{iPS} \Delta \zeta_{iPS} \approx n_{pPS} (\Delta x)^2 \end{array} \right. \Rightarrow \quad (6.1)$$

$$\Rightarrow \bar{C}_{iPS} \approx \frac{\bar{Q}_{PS}}{\left( -\underline{n}_{inlet} \cdot \bar{\underline{u}}_{iPS} \right) n_{pPS} (\Delta x)^2}, \quad \bar{Q}_{PS,eq} = \bar{Q}_{PS}$$

The subscript ‘ $i_{PS}$ ’ denotes  $i^{\text{th}}$  position associated with the PS in the 2D time-independent emission pattern of the inlet SPH particles and  $n_{pPS}$  is the number of such positions. If no point is detected as internal to the PS volume, the closest position in the pattern is selected. All the particles emitted by the points defined above are called PS inlet particles. It is the inlet particles which cross a PS at some time during their travel. The approximation symbols in (6.1) are due to considering null concentration fluctuations at source. The particle volume is locally assumed uniform and isotropic.

If the PS lies downstream from the inlet section, the following procedures integrate the model above. The numerical volume of the PS is extruded from the PS 2D surface to the inlet section and cannot overlap with other numerical volumes of PSs. The inlet section is defined to be almost parallel to the plane containing the 2D PS surface. SPH Cm-BE is frozen within the inlet region. Downstream, SPH Cm-BE is frozen if the particle flight time  $t_{FLY}$  (s) is smaller than the maximum frozen flight time  $t_{FLY,FRO,max}$  (s). This is defined to make the PS particles hold null  $\bar{C}$  until reaching the physical PS:

$$t_{FLY,FRO,max,PS} = \frac{d_{PS,inlet}}{\overline{u}_{inlet,\eta_{PS},\zeta_{PS}}}, \quad d_{PS,inlet} = d_{PS,PHYinlet} + L_{inlet}, \quad L_{inlet} = \left[ \text{int} \left( 4 \frac{h}{\Delta x} \right) + 1 \right] \Delta x + \frac{\Delta x}{2} \quad (6.2)$$

The velocity value in (6.2) is approximated by the corresponding time-dependent value of the first PS position detected by the code in the time-independent pattern of the inlet particles. The distance between the physical PS and the most upstream layer of inlet particles  $d_{PS,inlet}$  (m) is the distance between the physical PS and the physical inlet section ‘ $PHYinlet$ ’, increased of the length of the inlet region  $L_{inlet}$  (m). This is approximately equal to  $4h$  plus half the particle size. The physical PS has to be close enough to the inlet section to satisfy the following conditions: no other physical PS affects the flow upstream from the current PS; the flow locally satisfies Taylor’s translation hypothesis for turbulence. The latter considers that the accumulated effect of the stream-wise relative dispersion is negligible and that the mean flow is locally almost 1D. Under these conditions, the local mean velocity at source can be approximately replaced by the inlet mean velocity at the same inlet local coordinates, for the assessments of  $\bar{C}_{iPS}$  in (6.1) and  $t_{FLY,FRO,max}$  in (6.2). The mean velocity of a PS particle is deemed almost uniform along its trajectory between the numerical inlet section and the physical PS. Any pollutant-mass flux between a computational particle and a neighbouring particle having SPH Cm-BE frozen is inhibited to conserve the pollutant mass. The PS numerical geometry and  $t_{FLY,FRO,max}$  are optimised to reduce the distance between the barycentres of the numerical equivalent PS and of the physical PS. A convergence criterion for each distance component is  $2\Delta x$ . Multiple PSs are treated. Volume PSs are permitted, provided that they belong to the physical inlet section and only extend in the inlet region up to the numerical inlet section. Volume PSs have null  $t_{FLY,FRO,max}$ . The simulated PSs are stationary. ICs and the inlet BCs can impose additional time-dependent background concentrations.

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